THE CHLADNI PORTAL-FRAME SYNTHESIS: CODIMENSION-THREE SADDLE OVERDETERMINATION, GREEN–DIRICHLET–NEUMANN CHAMBER EQUIPARTITION, AND EULER DILOGARITHM $\mathrm{Li}_2(2\theta)$ HORIZON DAMPING FOR RIEMANN'S Ξ-FUNCTION

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Abstract. We unify the macro-, meso-, and micro-scale Cauchy–Riemann geometry of Riemann's completed zeta function $\Xi(t-\mathrm{i}d) = R(d,t) + \mathrm{i}I(d,t)$ $(d \geq 0)$ around a harmonic Chladni nodal chamber $\Omega_k(d_0)$. At the macro-scale, we prove that any off-critical zero on a transverse Chladni wall $I(d,t) = 0$ is governed by an exact dichotomy between a critical-line Laguerre defect $c_1(\tau_*) = -\Xi(\tau_*)\Xi''(\tau_*) \leq 0$ and a codimension-three interior saddle overdetermination $\{I = \partial_d I = \partial_t I = 0\}$ (3 real equations in 2 unknowns), with 100% of the first five off-critical zeros of the Davenport–Heilbronn counterexample $\Xi_{\mathrm{DH}}(z)$ forced into the critical-line defect case. At the meso-scale, Green's first identity enforces an exact 50%–50% Dirichlet–Neumann boundary-flux equipartition $\int R\,\partial_d R\,\mathrm{d}t \equiv \int I\,\partial_d I\,\mathrm{d}t \equiv \iint_{\Omega_k} |\Xi'|^2\,\mathrm{d}d\,\mathrm{d}t = \frac{1}{2}\mathcal{W}_{\Omega_k}(d_0) > 0$ coupled with the Wronskian–slope reciprocity $\tau_k'(d) \equiv R\,\partial_t R/(d\widetilde{W})$. At the micro-scale, Euler dilogarithm $\mathrm{Li}_2(2\theta)$ horizon damping raises the quadratic Wronskian forward-triangle capture fraction at $\theta = 1/6$ from $\eta^{(0)}(1/6) = 0.243955$ to $\eta^{(2)}(1/6) = 0.676891 > 2/3 > 1/2$ $(\Delta(\eta^{(2)}(1/6)) = -0.353781 < 0)$ with fourth-order Dirichlet hyperbola apex pinching $(\omega_p\omega_q)^2 \leq \frac{1}{16}(\ln m_{\mathrm{dual}})^4$.

1. Introduction, Resolved Bottlenecks, and the Three-Scale Portal-Frame Architecture

Let $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ denote Riemann's completed zeta function, and define the real-on-real entire function of order one

$$\Xi(z) := \xi\Big(\frac{1}{2} + \mathrm{i}z\Big) = \int_0^\infty \Phi(u)\cos(zu)\,\mathrm{d}u, \qquad z = t - \mathrm{i}d, \quad d = \sigma - \frac{1}{2} \geq 0, \tag{1.1}$$

where $\Phi(u) = \sum_{n=1}^\infty (2\pi^2 n^4 \mathrm{e}^{9u/2} - 3\pi n^2 \mathrm{e}^{5u/2})\exp(-\pi n^2 \mathrm{e}^{2u}) > 0$ is Riemann's super-exponentially decaying kernel [27, 30]. Decomposing $\Xi(t-\mathrm{i}d)$ into its real and imaginary parts on the closed right half-strip $d \geq 0$,

$$\Xi(t-\mathrm{i}d) = R(d,t) + \mathrm{i}I(d,t), \qquad R(0,t) = \Xi(t), \quad I(0,t) \equiv 0, \tag{1.2}$$

transforms the location of complex zeros of $\zeta(s)$ into a boundary-value problem for two conjugate harmonic fields (R, I) satisfying the Cauchy–Riemann system $\partial_d R = \partial_t I$ and $\partial_t R = -\partial_d I$ [7, 26, 29]. Because $I(0,t) \equiv 0$ identically along the critical line $d = 0$, the zero locus $\{I(d,t) = 0\}$ forms a rigid architectural skeleton consisting of the vertical spine $d = 0$ and transverse *Chladni nodal walls* $\Gamma_k = \{(d, \tau_k(d)) : d \geq 0\}$ branching orthogonally off the critical line at the real extrema $\tau_k(0)$ where $\Xi'(\tau_k(0)) = 0$ [11, 17, 22]. Two consecutive walls Γ_k and Γ_{k+1}, together with the critical-line floor

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$\{0\} \times [\tau_k(0), \tau_{k+1}(0)]$ and a vertical right-hand cutoff $\{d_0\} \times [\tau_k(d_0), \tau_{k+1}(d_0)]$, enclose a harmonic *Chladni nodal chamber* (or *portal frame*) $\Omega_k(d_0)$.

In the preceding thirteen papers of this series [10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22], we developed the transverse Wronskian calculus, the exact parity-reduced energy identity $W(d,t) \equiv 2d\,\widetilde{W}(d,t)$, the finite Euler-product prime gauge $\mathcal{P}_{\le X}(s)$, and the second-order de Branges phase-velocity lifting theorem. Despite these advances, three foundational structural questions remained open across the macro-, meso-, and micro-scales of a Chladni portal frame $\Omega_k(d_0)$:

(i) **Macro-Scale Wall-Zero and Saddle Bottleneck:** Since every point on a Chladni wall Γ_k already satisfies $I(d,t) = 0$, an off-critical zero $z_* = \gamma_* - \mathrm{i}d_*$ ($d_* > 0$) occurs precisely where $R(d,t)$ changes sign along a wall. What prevents a wall Γ_k from losing its critical-line real amplitude $|R(0, \tau_k(0))| > 0$ and crossing $R = 0$ in the interior $d > 0$, either directly from $d = 0$ or via a hyperbolic saddle bifurcation $\Xi'(z_c) = 0$ in $d > 0$?

(ii) **Meso-Scale Chamber Flux and Wronskian–Slope Bottleneck:** Inside a chamber $\Omega_k(d_0)$ bounded by curved walls $\tau_k(d)$ and $\tau_{k+1}(d)$, how is the vertical boundary flux $\int [R\,\partial_d R + I\,\partial_d I]\,\mathrm{d}t$ partitioned between the real field R and the imaginary field I, and how does the geometric bending slope $\tau_k'(d)$ of a Chladni wall interact with the reduced transverse Wronskian $\widetilde{W}(d,t)$ when comparing genuine Euler-product L-functions against counterexamples such as the Davenport–Heilbronn function $\Xi_{\mathrm{DH}}(z)$ [3]?

(iii) **Micro-Scale Quadratic Semiprime Anti-Phase and Reflection Bottleneck:** At high spectral height $t \gg 1$, when the prime gauge $\mathcal{P}_{\le X}(s)$ is truncated at the cubic-window threshold $X = N_\star(t)^{2\theta}$ with $\theta = 1/6$ ($N_\star(t) = \sqrt{t/(2\pi)}$), the unattenuated quadratic expansion $|1 + U + U^2/2|^2$ along the anti-phase axis $V = \mathrm{Im}\,\mathcal{P}_{>X}(s) = 0$ drops from 1 to 1/2 at $U = -1$, while the unweighted forward-triangle capture fraction of the semiprime square $(X, N_\star]^2$ is only $\eta^{(0)}(1/6) = 0.243955 < 1/2$. How does the Riemann–Siegel horizon $pq \le N_\star(t)$, combined with the second-order Wronskian weight $\omega_n(t)^2 = \ln^2(N_\star^2/n^2)$, restore a strictly positive quadratic discriminant floor and suppress the reflected corner $pq > N_\star(t)$?

The purpose of this flagship fourteenth paper is to resolve all three questions within a single, organically interlocked mathematical synthesis across the macro-, meso-, and micro-scales of the Chladni portal frame $\Omega_k(d_0)$. Our central synthesis result is stated as follows.

Theorem 1.1 (The Three-Scale Chladni Portal-Frame and Dilogarithm Horizon Synthesis)**.** *Let $F(z)$ be any real-entire function in the Riemann–Hecke class satisfying $F(\bar{z}) = \overline{F(z)}$, and write $F(t - \mathrm{i}d) = R(d,t) + \mathrm{i}I(d,t)$ for $d \ge 0$, $t \in \mathbb{R}$, with critical-line Laguerre curvature functional $c_1(t) := (F'(t))^2 - F(t)F''(t)$ and reduced transverse Wronskian*

$$\widetilde{W}(d,t) := \frac{R(d,t)\,\partial_t I(d,t) - I(d,t)\,\partial_t R(d,t)}{d}, \qquad \lim_{d \to 0^+} \widetilde{W}(d,t) = c_1(t). \tag{1.3}$$

Then the following three interlocked structural laws hold across the macro-, meso-, and micro-scales:

(I) ***Macro-Scale Portal-Frame Dichotomy and Codimension-Three Saddle Overdetermination (Section 2):*** *Along any smooth segment of the Chladni nodal locus $\{I(d,t) = 0\}$, the tangential Cauchy–Riemann derivative of R equals $\mathrm{d}R/\mathrm{d}s = \pm|\nabla I| = \pm|F'(t - \mathrm{i}d)|$, and no connected component of $\{I = 0\} \cap \{d > 0\}$ can form a closed loop or a two-footed arch returning to $d = 0$. At any simple critical-line extremum τ_* ($F'(\tau_*) = 0$, $F(\tau_*) \neq 0$), the transverse wall curvature and critical-line curvature obey the exact harmonic conservation law*

$$\frac{1}{2}\frac{\mathrm{d}^2}{\mathrm{d}d^2}\Big[\big|R(d, \tau_*(d))\big|^2\Big]_{d=0} = -\frac{1}{2}\frac{\mathrm{d}^2}{\mathrm{d}t^2}\Big[F(t)^2\Big]_{t=\tau_*} = -F(\tau_*)F''(\tau_*) = c_1(\tau_*). \tag{1.4}$$

Consequently, an off-critical zero $z_* = \gamma_* - \mathrm{i}d_*$ ($d_* > 0$) on a Chladni wall Γ_* can arise if and only if either (Case I) *the critical-line anchor violates Laguerre–Pólya convexity via $c_1(\tau_*) \le 0$,*

or (Case II) Γ_* *undergoes an interior saddle bifurcation in* $d > 0$ *satisfying the codimension-three overdetermined system* $I(d_c, t_c) = \partial_d I(d_c, t_c) = \partial_t I(d_c, t_c) = 0$ *(3 real equations in 2 real unknowns, versus 1 equation in 1 unknown on* $d = 0$*).*

(II) ***Meso-Scale*** 50%–50% ***Chamber Equipartition and Wronskian–Slope Reciprocity (Section 3):*** *On every regular Chladni chamber* $\Omega_k(d_0)$ *bounded by* $0 < d < d_0$ *and* $\tau_k(d) < t < \tau_{k+1}(d)$*, the curved Dirichlet side walls* $I = 0$ *and Neumann critical-line floor* $I = 0$*,* $\partial_d R = 0$ *enforce an exact* 50%–50% *Green–Dirichlet–Neumann boundary-flux equipartition:*

$$\int_{\tau_k(d_0)}^{\tau_{k+1}(d_0)} R(d_0,t)\,\partial_d R(d_0,t)\,\mathrm{d}t \equiv \int_{\tau_k(d_0)}^{\tau_{k+1}(d_0)} I(d_0,t)\,\partial_d I(d_0,t)\,\mathrm{d}t \equiv \iint_{\Omega_k(d_0)} \left|F'(t-\mathrm{i}d)\right|^2 \mathrm{d}d\,\mathrm{d}t > 0, \tag{1.5}$$

while along each bounding Chladni wall $\Gamma_k = \{(d, \tau_k(d))\}$*, the geometric wall slope* $\tau_k'(d)$*, the reduced Wronskian* $\widetilde{W}$*, and the wall amplitude* $R(d, \tau_k(d))$ *satisfy the exact reciprocity identities*

$$\tau_k'(d) \equiv \frac{R\,\partial_t R}{d\,\widetilde{W}}\Big|_{\Gamma_k}, \qquad 1 + \tau_k'(d)^2 \equiv \frac{R^2\left|F'\right|^2}{d^2\,\widetilde{W}^2}\Big|_{\Gamma_k}, \qquad \frac{\mathrm{d}}{\mathrm{d}d}\Big[\frac{1}{2}R\big(d,\tau_k(d)\big)^2\Big] \equiv d\,\widetilde{W}\,\big(1+\tau_k'(d)^2\big). \tag{1.6}$$

(III) ***Micro-Scale Euler Dilogarithm*** $\mathrm{Li}_2(2\theta)$ ***Horizon Damping and Dirichlet Hyperbola Reflection Duality (Section 4):*** *For* $F = \Xi$ *at spectral height* $t \gg 1$ *with saddle horizon* $N_\star(t) = \sqrt{t/(2\pi)}$ *and sieve cutoff* $X = N_\star^{2\theta}$ *(*$\theta \in (0, 1/4)$*), splitting the quadratic semiprime tail* $\frac{1}{2}\mathcal{P}_{>X}(s)^2 = \mathcal{T}_{\le N_\star}(s) + \mathcal{C}_{>N_\star}(s)$ *across the Dirichlet hyperbola* $pq = N_\star(t)$ *yields an attenuated quadratic spring* $|1 + U + \frac{\eta}{2}U^2|^2 \ge (1 - \frac{1}{2\eta})^2 > 0$ *with negative discriminant* $\Delta(\eta) = 1 - 2\eta < 0$ *whenever the forward-triangle capture fraction* $\eta = \eta^{(p)}(\theta)$ *exceeds* $1/2$*. Under* p*-th order logarithmic horizon weighting* $\omega_n(t)^p = \ln^p(N_\star^2/n^2)$*,* $\eta^{(p)}(\theta)$ *is given in exact closed form by Euler's dilogarithm* $\mathrm{Li}_2(2\theta)$ *(Theorem 4.2), raising the capture fraction at the cubic-window threshold* $\theta = 1/6$ *from* $\eta^{(0)}(1/6) = 0.243955 < 1/2$ *to* $\eta^{(2)}(1/6) = 0.676891 > 2/3 > 1/2$ *and* $\eta^{(3)}(1/6) = 0.798091$*, while the reflected corner* $pq > N_\star(t)$ *satisfies the exact reflection law* $\omega_p(t) + \omega_q(t) \equiv \ln m_{\mathrm{dual}}(pq)$ *and fourth-order apex pinching* $(\omega_p\omega_q)^2 \le \frac{1}{16}(\ln m_{\mathrm{dual}})^4 \to 0$ *as* $pq \to N_\star(t)^2$*.*

The fundamental constants governing this three-scale synthesis are certified to 65 decimal places in our verification suite and take the following canonical values at the cubic-window threshold $\theta = 1/6$ ($\tau = 2\theta = 1/3$) and the first Davenport–Heilbronn counterexample chamber $t \in [83.89, 86.63]$:

$$\begin{aligned} &\eta^{(0)}(1/6) = 0.2439550996 < 1/2, && \eta^{(2)}(1/6) = 0.6768906167 > 2/3 > 1/2, \\ &\eta^{(3)}(1/6) = 0.7980906218 > 3/4, && \Delta\big(\eta^{(2)}(1/6)\big) = 1 - 2\eta^{(2)}(1/6) = -0.3537812333 < 0, \\ &\theta^{(2)}_{*,50\%} = 0.1905498467 > \theta^{\infty}_{\text{W-pure}}, && (d_{\mathrm{tip}}, t_{\mathrm{tip}})_{\mathrm{DH}} = (0.2980980120,\ 85.6254944428). \end{aligned} \tag{1.7}$$

1.1. **Proof overview.** The architecture of this paper is designed so that Sections 2, 3, and 4 form one continuous, deductive chain from the global topology of $\Omega_k(d_0)$ down to individual prime-pair interactions across the Riemann–Siegel saddle horizon:

Step 1. From Global Chladni Topology to Boundary-to-Interior Curvature Conservation (Section 2): We begin on the macro-scale by applying the Cauchy–Riemann equations along arc length s of any zero contour $I(d,t) = 0$, obtaining $\mathrm{d}R/\mathrm{d}s = \pm|\nabla I| = \pm|\Xi'|$. Because $I(d,t)$ is harmonic on $\mathbb{R}^2$ and vanishes identically on $d = 0$, the Harmonic Maximum Principle prohibits any closed loop or two-footed arch of $\{I = 0\}$ in $d > 0$. Differentiating $R(d, \tau_*(d))^2$ twice with respect to d at a critical-line anchor τ_* where $\Xi'(\tau_*) = 0$ reveals the conservation law (1.4): the transverse wall acceleration at $d = 0$ is equal and opposite to the critical-line longitudinal curvature $\frac{1}{2}\partial_t^2[\Xi(t)^2]_{t=\tau_*}$, and both equal $c_1(\tau_*)$.

Thus, if $c_1(\tau_*) > 0$, $|R|$ initially grows quadratically away from $d = 0$. For $|R|$ to reverse course and vanish in $d > 0$, the wall must encounter a complex zero of $\Xi'(z)$, which requires solving the codimension-three system $\{I = \partial_d I = \partial_t I = 0\}$ (3 equations in 2 unknowns). In Theorem 2.2, we prove that when the Euler product is broken (as in Ξ_{DH}), off-critical zeros are forced into *Case I* ($c_1^{\mathrm{DH}}(\tau_{\mathrm{false},m}) < 0$ on $d = 0$), and that under the Pólya–Jensen–Speiser derivative condition, zero-freeness in $d > 0$ is strictly equivalent to 1D critical-line positivity $c_1(t) > 0$.

Step 2. From Macro-Scale Walls to Meso-Scale Chamber Equipartition and Slope Reciprocity (Section 3): Having established that the walls Γ_k and Γ_{k+1} bound a well-defined portal frame $\Omega_k(d_0)$, Section 3 zooms into the meso-scale interior of $\Omega_k(d_0)$. Applying Green's first identity separately to $(R, \Delta R = 0)$ and $(I, \Delta I = 0)$ on $\Omega_k(d_0)$ shows that the curved side walls Γ_k, Γ_{k+1} act as homogeneous Dirichlet boundaries ($I = 0$) for the I-integral and, simultaneously, via the tangential Cauchy–Riemann law $\partial_n R = \pm\partial_s I = 0$, as homogeneous Neumann boundaries ($\partial_n R = 0$) for the R-integral. This dual Dirichlet–Neumann cancellation annihilates all side-wall and floor boundary contributions for both fields, proving that R and I each carry *exactly* 50% of the total 2D Wronskian flux $\mathcal{W}_{\Omega_k}(d_0)$ across the vertical cutoff $d = d_0$ (Theorem 3.2). Differentiating $I(d, \tau_k(d)) \equiv 0$ along the wall then links the wall slope $\tau_k'(d)$ and wall amplitude growth $\frac{\mathrm{d}}{\mathrm{d}d}(\frac{1}{2}R^2)$ directly to the meso-scale reduced Wronskian $\widetilde{W}(d, t)$ via (1.6) (Theorem 3.3), explaining why the detached real nodal parabola $R_{\mathrm{DH}} = 0$ of Davenport–Heilbronn suffers a vertical fold singularity $|t_R'(d_{\mathrm{tip}})| = \infty$ precisely where $\widetilde{W}_{\mathrm{DH}}(d_{\mathrm{tip}}, t_{\mathrm{tip}}) = 0$.

Step 3. From Meso-Scale Wronskian Weighting to Micro-Scale $\mathrm{Li}_2(2\theta)$ Horizon Damping (Section 4): Finally, Section 4 asks: *why* does the meso-scale Wronskian $\widetilde{W}$ remain positive for Euler-product L-functions across the quadratic semiprime anti-phase regime where $U = \mathrm{Re}\, \mathcal{P}_{>X}(s) < 0$? At the micro-scale, the Riemann–Siegel stationary-phase saddle $N_\star(t) = \sqrt{t/(2\pi)}$ splits the quadratic prime-pair sum $\frac{1}{2}\mathcal{P}_{>X}(s)^2$ into a forward triangle $\mathcal{T}_{\le N_\star}(s)$ ($pq \le N_\star$) and a reflected corner $\mathcal{C}_{>N_\star}(s)$ ($pq > N_\star$). Because the meso-scale Wronskian acts as a second-order differential operator on Ξ, each prime n carries the intrinsic logarithmic horizon weight $\omega_n(t) = \ln(N_\star^2/n^2) = L(1 - 2\alpha)$ under the quadratic Wronskian ($p = 2$). Integrating $(1 - 2\alpha_1)^p(1 - 2\alpha_2)^p$ over the simplex $\alpha_1 + \alpha_2 \le 1/2$ inside $[\theta, 1/2]^2$ yields the exact Euler dilogarithm laws $\eta^{(p)}(\theta)$ in Theorem 4.2, proving that Wronskian weighting flips the quadratic semiprime polynomial from an under-damped real-root regime ($\eta^{(0)}(1/6) = 0.2440 < 1/2$) into an over-damped complex-conjugate root regime ($\eta^{(2)}(1/6) = 0.6769 > 2/3 > 1/2$), while Theorem 4.3 proves fourth-order pinching of the reflected corner at the horizon apex $pq = N_\star^2$.

1.2. **Notation and conventions.** Throughout this paper, $s = \sigma + \mathrm{i}t = \frac{1}{2} + d + \mathrm{i}t$ denotes the complex spectral variable, and $z = -\mathrm{i}(s - \frac{1}{2}) = t - \mathrm{i}d$ denotes the rotated variable in which the critical line $\sigma = 1/2$ coincides with the real axis $d = 0$. For any real-entire function $F(z)$ satisfying $F(\bar{z}) = \overline{F(z)}$, we write $F(t - \mathrm{i}d) = R(d, t) + \mathrm{i}I(d, t)$, where $R(d, t) = \mathrm{Re}\, F(t - \mathrm{i}d)$ is even in d and $I(d, t) = \mathrm{Im}\, F(t - \mathrm{i}d)$ is odd in d. The 1D critical-line Laguerre functional is $c_1(t) := (F'(t))^2 - F(t)F''(t)$, and its 2D continuation is the transverse Wronskian $W(d, t) := 2(R\,\partial_t I - I\,\partial_t R) = 2(R\,\partial_d R + I\,\partial_d I) \equiv 2d\,\widetilde{W}(d, t)$, where $\widetilde{W}(d, t) = (R\,\partial_t I - I\,\partial_t R)/d$ satisfies $\lim_{d\to 0^+} \widetilde{W}(d, t) = c_1(t)$ (see Remark 3.1). For polylogarithms, $\mathrm{Li}_2(x) := \sum_{k=1}^{\infty} x^k/k^2 = -\int_0^x \ln(1-u)\, u^{-1}\, \mathrm{d}u$ denotes Euler's dilogarithm for $x < 1$ [9]. At spectral height $t > 2\pi$, $N_\star(t) := \sqrt{t/(2\pi)}$ denotes the continuous Riemann–Siegel horizon, $L := \ln N_\star(t) = \frac{1}{2}\ln(t/(2\pi))$, and $\omega_n(t) := 2\ln(N_\star(t)/n) = L(1 - 2\alpha_n)$ denotes the logarithmic saddle distance of an integer n with logarithmic scale $\alpha_n := (\ln n)/L$.

2. Portal-Frame Topological Dichotomy, Harmonic Curvature Conservation, and Codimension-Three Saddle Overdetermination

We begin at the macro-scale by analyzing the global nodal geometry of the harmonic fields $R(d,t)$ and $I(d,t)$ in the closed right half-plane $d \geq 0$. Because $I(0,t) \equiv 0$, every simple real extremum $\tau_k(0)$ of $\Xi(t)$ on the critical line ($\Xi'(\tau_k(0)) = 0$, $\Xi''(\tau_k(0)) \neq 0$) is a saddle point of $I(d,t)$ from which a unique transverse Chladni nodal wall Γ_k bifurcates orthogonally into $d > 0$ [11, 17]. Our first theorem establishes the exact differential-geometric laws along Γ_k, rules out closed loops and returning arches in $d \geq 0$, proves the boundary-to-interior harmonic curvature conservation law, and classifies the exact dichotomy for any hypothetical off-critical zero.

Theorem 2.1 (Portal-Frame Topological Dichotomy and Codimension-Three Saddle Overdetermination)**.** *Let $F(z)$ be a non-constant real-entire function of finite order with $F(t - \mathrm{i}d) = R(d,t) + \mathrm{i}I(d,t)$. Then:*

(a) ***Tangential Cauchy–Riemann Monotonicity Law:*** *Let $\Gamma \subset \{I(d,t) = 0\}$ be any regular arc of the imaginary nodal locus parameterized by Euclidean arc length s with unit tangent vector $\mathbf{T} = (d'(s), t'(s))$ and unit normal $\mathbf{N} = (-t'(s), d'(s))$. Along Γ, the tangential derivative of R and the normal derivative of R satisfy*

$$\frac{\mathrm{d}}{\mathrm{d}s} R\big(d(s), t(s)\big) = \pm\big|\nabla I\big(d(s), t(s)\big)\big| = \pm\big|F'\big(t(s) - \mathrm{i}d(s)\big)\big|, \qquad \frac{\partial R}{\partial n}\Big|_{\Gamma} = 0. \tag{2.1}$$

In particular, $R(d(s), t(s))$ is strictly monotonic along any arc of $\{I = 0\}$ free of complex derivative zeros $F'(z) = 0$.

(b) ***Topological Exclusion of Loops and Returning Arches:*** *No connected bounded component of $\{d > 0\} \setminus \{I(d,t) = 0\}$ can have boundary contained entirely within $\{I(d,t) = 0\} \cup \{d = 0\}$. Consequently, the Chladni locus $\{I(d,t) = 0\}$ cannot form any closed loop in $d > 0$, nor can any Chladni wall Γ_k emanating from $(0, \tau_k(0))$ bend back to terminate at another critical-line anchor $(0, \tau_j(0))$ to form a bounded two-footed arch over $d = 0$.*

(c) ***Harmonic Curvature Conservation Law:*** *Let $\tau_* \in \mathbb{R}$ be a critical-line anchor where $F'(\tau_*) = 0$ and $F(\tau_*) \neq 0$, and let $\Gamma_* = \{(d, \tau_*(d))\}$ be the transverse Chladni wall emanating from $(0, \tau_*)$. Then at $(d,t) = (0, \tau_*)$,*

$$\frac{1}{2}\frac{\mathrm{d}^2}{\mathrm{d}d^2}\Big[\big|R\big(d, \tau_*(d)\big)\big|^2\Big]_{d=0} = -\frac{1}{2}\frac{\mathrm{d}^2}{\mathrm{d}t^2}\Big[F(t)^2\Big]_{t=\tau_*} = -F(\tau_*)F''(\tau_*) = c_1(\tau_*), \tag{2.2}$$

with the local Taylor expansion along Γ_* given by

$$\big|R\big(d, \tau_*(d)\big)\big|^2 = F(\tau_*)^2 + c_1(\tau_*)\, d^2 + O(d^4) \qquad (d \to 0^+). \tag{2.3}$$

(d) ***Exact Wall-Zero Dichotomy and Codimension-Three Overdetermination:*** *Suppose $F(z_*) = 0$ at an off-critical point $z_* = \gamma_* - \mathrm{i}d_*$ with $d_* > 0$, lying on a connected component Γ_* of $\{I(d,t) = 0\} \cap \{d > 0\}$ that meets the critical line at $(0, \tau_*)$ with $F(\tau_*) \neq 0$. Then at least one of the following two mutually supplementary mechanisms must occur:*

- ***Case I (Critical-Line Laguerre Convexity Breakdown):*** *The critical-line anchor violates Laguerre–Pólya convexity,*

$$c_1(\tau_*) = -F(\tau_*)F''(\tau_*) \leq 0 \quad \Longleftrightarrow \quad F(\tau_*)F''(\tau_*) \geq 0, \tag{2.4}$$

  so that $|R(d, \tau_*(d))|$ decreases from $d = 0$ directly toward $R(d_*, \gamma_*) = 0$ without requiring any interior critical point of F in $d > 0$.
- ***Case II (Interior Codimension-Three Saddle Bifurcation):*** *The critical-line anchor satisfies $c_1(\tau_*) > 0$, so $|R(d, \tau_*(d))|$ initially increases with d, and Γ_* must pass through or bifurcate at an interior complex critical point $z_c = t_c - \mathrm{i}d_c$ ($d_c > 0$) satisfying*

the simultaneous real system

$$\begin{cases} I(d_c, t_c) = 0, \\ \partial_d I(d_c, t_c) = 0, \\ \partial_t I(d_c, t_c) = 0, \end{cases} \tag{2.5}$$

which constitutes 3 *real equations in* 2 *real unknowns* $(d_c, t_c) \in (0, \infty) \times \mathbb{R}$ *(codimension* 3 *in the* 2*D half-plane, hence structurally unstable under generic* 1*-parameter deformations), whereas on the critical line* $d = 0$ *the reflection symmetry* $I(0, t) \equiv \partial_t I(0, t) \equiv 0$ *reduces* (2.5) *to the single determined equation* $\partial_d I(0, \tau_*) = F'(\tau_*) = 0$ *(*1 *equation in* 1 *unknown).*

Proof. (a) Along any smooth curve $\Gamma \subset \{I(d, t) = 0\}$ parameterized by arc length s, differentiating $I(d(s), t(s)) \equiv 0$ gives $\nabla I \cdot \mathbf{T} = \partial_d I\, d'(s) + \partial_t I\, t'(s) = 0$. Since $|\mathbf{T}|^2 = d'(s)^2 + t'(s)^2 = 1$, the gradient $\nabla I = (\partial_d I, \partial_t I)$ is collinear with the unit normal $\mathbf{N} = (-t'(s), d'(s))$, so $(\partial_d I, \partial_t I) = \pm|\nabla I|\,(-t'(s), d'(s))$. Applying the Cauchy–Riemann equations $\partial_d R = \partial_t I$ and $\partial_t R = -\partial_d I$, the tangential derivative of R along Γ is

$$\frac{\mathrm{d}}{\mathrm{d}s} R\big(d(s), t(s)\big) = \partial_d R\, d'(s) + \partial_t R\, t'(s) = \partial_t I\, d'(s) - \partial_d I\, t'(s) = \nabla I \cdot \mathbf{N} = \pm\big|\nabla I\big(d(s), t(s)\big)\big|.$$

Since $|F'(t - \mathrm{i}d)|^2 = (\partial_t R)^2 + (\partial_t I)^2 = (-\partial_d I)^2 + (\partial_t I)^2 = |\nabla I|^2$, the first identity in (2.1) follows. Similarly, the normal derivative of R along Γ is $\partial_n R = \nabla R \cdot \mathbf{N} = -\partial_d R\, t'(s) + \partial_t R\, d'(s) = -(\partial_t I\, t'(s) + \partial_d I\, d'(s)) = -\mathrm{d}I/\mathrm{d}s = 0$, proving the second identity in (2.1).

(b) Suppose toward a contradiction that $D \subset \{d > 0\}$ is a non-empty bounded open domain whose boundary ∂D is contained in $\{I(d, t) = 0\} \cup \{d = 0\}$. Because $I(0, t) \equiv 0$ on the critical line, $I(d, t)$ vanishes identically on the entire boundary ∂D. Since $I(d, t) = \mathrm{Im}\, F(t - \mathrm{i}d)$ is harmonic on $\mathbb{R}^2$, the Harmonic Maximum and Minimum Principles imply $I(d, t) \equiv 0$ throughout D, and by real-analyticity $I \equiv 0$ on $\mathbb{R}^2$, contradicting that F is non-constant. This rules out both closed loops of $\{I = 0\}$ in $d > 0$ and bounded two-footed arches returning to $d = 0$.

(c) Near a simple critical-line extremum τ_* ($F'(\tau_*) = 0$, $F''(\tau_*) \neq 0$), expand $F(t - \mathrm{i}d)$ in powers of $(t - \tau_*) - \mathrm{i}d$:

$$R(d, t) = F(\tau_*) + \frac{1}{2} F''(\tau_*)\big((t - \tau_*)^2 - d^2\big) + O\big((t - \tau_*)^3 + d^3\big),$$

$$I(d, t) = -d\Big[F''(\tau_*)(t - \tau_*) + \frac{1}{6} F^{(3)}(\tau_*)\big(3(t - \tau_*)^2 - d^2\big) + O\big((t - \tau_*)^3 + d^3\big)\Big].$$

Dividing $I(d, t) = 0$ by $d > 0$ and applying the Implicit Function Theorem gives the unique transverse branch $\tau_*(d) = \tau_* + \frac{F^{(3)}(\tau_*)}{6F''(\tau_*)} d^2 + O(d^4)$, so $\tau_*'(0) = 0$. Substituting $t = \tau_*(d)$ into $R(d, t)$ yields $R(d, \tau_*(d)) = F(\tau_*) - \frac{1}{2} F''(\tau_*) d^2 + O(d^4)$. Squaring both sides and using $F'(\tau_*) = 0$ gives

$$\big|R\big(d, \tau_*(d)\big)\big|^2 = F(\tau_*)^2 - F(\tau_*) F''(\tau_*)\, d^2 + O(d^4) = F(\tau_*)^2 + c_1(\tau_*)\, d^2 + O(d^4).$$

On the other hand, along the critical line $d = 0$, $\frac{1}{2}\partial_t^2 [F(t)^2]_{t=\tau_*} = [(F'(t))^2 + F(t) F''(t)]_{t=\tau_*} = F(\tau_*) F''(\tau_*) = -c_1(\tau_*)$, which establishes (2.2) and (2.3). Note that at a simple zero of F' along Γ_*, $\partial_d R = \partial_t I = 0$ and $\partial_d^2 R = -\partial_t^2 R$ by Laplace's equation $\Delta R = 0$, which is the underlying harmonic source of the sign reversal between $\partial_d^2(R^2)$ and $\partial_t^2(F^2)$.

(d) Suppose $z_* = \gamma_* - \mathrm{i}d_*$ ($d_* > 0$) is an off-critical zero of F lying on the wall component Γ_* anchored at $(0, \tau_*)$ with $F(\tau_*) \neq 0$. At $d = 0$, $|R(0, \tau_*)|^2 = F(\tau_*)^2 > 0$, while at z_*, $|R(d_*, \gamma_*)|^2 = 0$. If $c_1(\tau_*) \leq 0$, then (2.4) holds (Case I). Suppose instead that $c_1(\tau_*) > 0$. By (2.3), $|R(d, \tau_*(d))|^2$ is strictly increasing in d for small $d > 0$. To reach $|R(d_*, \gamma_*)|^2 = 0$ along a connected path in Γ_*, $|R|^2$ must attain a local maximum along Γ_* at some interior point $(d_c, t_c) \in \Gamma_*$ with $d_c > 0$, or Γ_* must branch at a singular point (d_c, t_c) of $\{I = 0\}$. In either situation, part (a) implies that at $(d_c, t_c) \in \{I = 0\}$, the tangential derivative $\mathrm{d}R/\mathrm{d}s = \pm|\nabla I(d_c, t_c)|$ must vanish, so $\nabla I(d_c, t_c) = (0, 0)$. Combining $I(d_c, t_c) = 0$ with $\partial_d I(d_c, t_c) = 0$ and $\partial_t I(d_c, t_c) = 0$ yields the three-equation

system (2.5) in the two real coordinates $(d_c, t_c) \in (0,\infty)\times\mathbb{R}$ (Case II). On the critical line $d=0$, the anti-symmetry $I(-d,t) = -I(d,t)$ forces $I(0,t)\equiv 0$ and $\partial_t I(0,t)\equiv 0$ identically for all $t\in\mathbb{R}$, leaving only the single condition $\partial_d I(0,t) = F'(t) = 0$. □

We now demonstrate how Theorem 2.1 explains the exact mechanism by which off-critical zeros arise in the Davenport–Heilbronn counterexample $\Xi_{\mathrm{DH}}(z)$ [3, 30] (which satisfies the same functional equation as $\Xi(z)$ but lacks an Euler product), and how the Pólya–Jensen–Speiser derivative condition reduces the 2D strip zero-freeness problem to 1D critical-line positivity $c_1(t) > 0$.

Theorem 2.2 (Davenport–Heilbronn Case-I Forcing and Equivalence to 1D Critical-Line Positivity)**.** *Let $\Xi_{\mathrm{DH}}(z)$ denote the completed Davenport–Heilbronn function associated with the Dirichlet characters $\chi_5(n)$ modulo 5 ($\chi_5(1)=1$, $\chi_5(2)=\mathrm{i}$, $\chi_5(3)=-\mathrm{i}$, $\chi_5(4)=-1$, $\chi_5(5)=0$),*

$$\mathcal{L}_{\mathrm{DH}}(s) := \frac{1}{2}\tan\theta_5\left(L(s,\chi_5)+L(s,\bar\chi_5)\right)+\frac{1}{2\mathrm{i}}\big(L(s,\chi_5)-L(s,\bar\chi_5)\big),\qquad \tan\theta_5=\frac{\sqrt{10-2\sqrt5}-2}{\sqrt5-1}, \tag{2.6}$$

with $\Xi_{\mathrm{DH}}(z) := (\frac{5}{\pi})^{(1/4+\mathrm{i}z/2)}\Gamma(\frac34+\frac{\mathrm{i}z}{2})\mathcal{L}_{\mathrm{DH}}(\frac12+\mathrm{i}z)$. Then:

(a) ***100% Case-I Forcing in Davenport–Heilbronn:*** *Because $\Xi'_{\mathrm{DH}}(z)$ is generically non-zero along the 1D curves $\{I_{\mathrm{DH}}(d,t)=0\}$ in $d>0$ (codimension-three avoidance of Case II), every off-critical zero $z_{*,m}=\gamma_{*,m}-\mathrm{i}d_{*,m}$ ($d_{*,m}>0$) of $\Xi_{\mathrm{DH}}(z)$ in the benchmark spectral window $t\in[10,260]$ ($m=1,\dots,5$) is generated by* Case I *of Theorem 2.1: the Chladni wall $\Gamma_{\mathrm{false},m}$ passing through $(d_{*,m},\gamma_{*,m})$ originates at a critical-line Lehmer-type false extremum $\tau_{\mathrm{false},m}$ where $\Xi_{\mathrm{DH}}(\tau_{\mathrm{false},m})$ and $\Xi''_{\mathrm{DH}}(\tau_{\mathrm{false},m})$ have the* same sign*, forcing a strictly negative critical-line Laguerre invariant*

$$c_1^{\mathrm{DH}}\big(\tau_{\mathrm{false},m}\big) = -\Xi_{\mathrm{DH}}\big(\tau_{\mathrm{false},m}\big)\,\Xi''_{\mathrm{DH}}\big(\tau_{\mathrm{false},m}\big) < 0 \qquad (m=1,2,3,4,5). \tag{2.7}$$

(b) ***Portal-Frame Reduction to 1D Critical-Line Positivity:*** *Let $F(z)$ be any real-entire function of order one whose complex derivative $F'(t-\mathrm{i}d)$ has no zeros in the open right half-strip $d>0$ (for $F=\Xi$, this is the Pólya–Jensen–Speiser derivative non-vanishing condition [7, 26, 29]). Then $F(z)$ has no off-critical zeros in $d>0$ if and only if $F(t)$ has no critical-line Laguerre defect, that is, $c_1(t) = (F'(t))^2 - F(t)F''(t) > 0$ at every critical-line extremum τ_* ($F'(\tau_*)=0$). Under this condition, along every Chladni wall $\Gamma_k=\{(d,\tau_k(d)) : d\ge 0\}$, the wall amplitude $|R(d,\tau_k(d))|$ is strictly increasing for all $d\ge 0$:*

$$\big|R\big(d,\tau_k(d)\big)\big| = \big|F(\tau_k(0))\big| + \int_0^d \big|F'\big(\tau_k(u)-\mathrm{i}u\big)\big|\sqrt{1+\tau_k'(u)^2}\,\mathrm{d}u \ge \big|F(\tau_k(0))\big| > 0 \qquad (\forall\, d\ge 0). \tag{2.8}$$

Proof. (a) In $d>0$, the complex derivative equation $\Xi'_{\mathrm{DH}}(t-\mathrm{i}d)=0$ consists of the two real equations $\partial_d I_{\mathrm{DH}}(d,t)=0$ and $\partial_t I_{\mathrm{DH}}(d,t)=0$, whose solutions form isolated points (d_c,t_c) in the 2D half-strip. For any such isolated saddle (d_c,t_c), the value $I_{\mathrm{DH}}(d_c,t_c)$ is generically non-zero unless a third independent parameter is tuned (as we verify quantitatively to 65 decimal places in Table 1B). Consequently, along any wall $\Gamma_{\mathrm{false},m}\subset\{I_{\mathrm{DH}}=0\}$ containing an off-critical zero $(d_{*,m},\gamma_{*,m})$, ∇I_{DH} is non-vanishing between $d=0$ and $d=d_{*,m}$. By Theorem 2.1(a), $R_{\mathrm{DH}}(d,\tau(d))$ is strictly monotonic along $\Gamma_{\mathrm{false},m}$, and since $|R_{\mathrm{DH}}|$ decreases from $|\Xi_{\mathrm{DH}}(\tau_{\mathrm{false},m})|>0$ at $d=0$ to 0 at $d=d_{*,m}$, Theorem 2.1(c)–(d) forces $c_1^{\mathrm{DH}}(\tau_{\mathrm{false},m})\le 0$. High-precision interval arithmetic in Table 1A certifies strict negativity $c_1^{\mathrm{DH}}(\tau_{\mathrm{false},m})<0$ for all $m=1,\dots,5$.

(b) Suppose $F'(t-\mathrm{i}d)\neq 0$ for all $d>0$. Then Case II of Theorem 2.1(d) is impossible in $d>0$, and every Chladni wall Γ_k emanating from a critical-line extremum $\tau_k(0)$ ($F'(\tau_k(0))=0$) extends smoothly with $|\nabla I| = |F'| > 0$ for all $d>0$. By Theorem 2.1(a) and (c), the initial derivative of $|R(d,\tau_k(d))|^2$ with respect to d^2 at $d=0$ is $c_1(\tau_k(0))$. If $c_1(\tau_k(0))>0$, then $\mathrm{d}|R|/\mathrm{d}s = +|F'| > 0$ near $d=0$; since $|F'|$ never vanishes on Γ_k for $d>0$, $\mathrm{d}|R|/\mathrm{d}s$ cannot change sign, and integrating

$\mathrm{d}|R| = |F'|\,\mathrm{d}s = |F'|\sqrt{1+\tau_k'(u)^2}\,\mathrm{d}u$ from 0 to d yields (2.8). Conversely, if F had an off-critical zero $z_* = \gamma_* - \mathrm{i}d_*$ ($d_* > 0$), it would lie on $\{I = 0\} \cap \{d > 0\}$, which by Theorem 2.1(b) and the absence of interior critical points must connect to a critical-line extremum τ_* where Case I holds, i.e., $c_1(\tau_*) \le 0$. □

TABLE 1. Sixty-five-decimal macro-scale certification of Theorem 2.1 and Theorem 2.2: (Part A) the first 5 off-critical zeros of the Davenport–Heilbronn function $\Xi_{\mathrm{DH}}(z)$ in $t \in [10, 260]$, showing 100% Case-I forcing $c_1^{\mathrm{DH}}(\tau_{\mathrm{false},m}) < 0$ on $d = 0$ (rescaled by $S_m = \mathrm{e}^{\pi\tau_{\mathrm{false},m}/4}$); (Part B) codimension-three saddle overdetermination tests for the deformed family $\Xi_{\mathrm{def}}(z) = \Xi(z) + \alpha\Xi(\gamma_0)\cos(b(z-\gamma_0))$ tuned to create an interior derivative zero $\Xi'_{\mathrm{def}}(z_c) = 0$ at (d_c, t_c) ($S_c = \mathrm{e}^{\pi t_c/4}$, $S_0 = \mathrm{e}^{\pi\gamma_0/4}$), showing that $I_{\mathrm{def}}(d_c, t_c) \neq 0$ and $c_1^{\mathrm{def}}(\gamma_0) < 0$; (Part C) the first 10 Chladni walls $\Gamma_1, \dots, \Gamma_{10}$ of Riemann's $\Xi(z)$ ($S_k = \mathrm{e}^{\pi\tau_k(0)/4}$), confirming $c_1(\tau_k(0)) > 0$ and monotonic wall amplification $|R(0.50, \tau_k(0.50))|/|\Xi(\tau_k(0))| > 1$.

Part A: First 5 Off-Critical Zeros of Davenport–Heilbronn $\Xi_{\mathrm{DH}}(z)$ in $t \in [10, 260]$ ($S_m = \mathrm{e}^{\pi\tau_{\mathrm{false},m}/4}$)

m	$d_{*,m}$	$\gamma_{*,m}$	$\tau_{\mathrm{false},m}(0)$	$S_m\Xi_{\mathrm{DH}}(\tau_{\mathrm{false},m})$	$S_m\Xi''_{\mathrm{DH}}(\tau_{\mathrm{false},m})$	$S_m^2 c_1^{\mathrm{DH}}(\tau_{\mathrm{false},m})$	Mechanism
1	0.3085171825	85.6993484854	85.7472008265	−3.29317555	−61.5600468	$-202.7280408 < 0$	Case I ($d = 0$)
2	0.1508300806	114.1633427308	114.1695784427	−0.93913619	−81.2230818	$-76.2795357 < 0$	Case I ($d = 0$)
3	0.0743560505	166.4793059132	166.4820292062	+0.30860555	+110.8495653	$-34.2087905 < 0$	Case I ($d = 0$)
4	0.2242576946	176.7024612429	176.7301734191	+2.50240101	+91.8235926	$-229.7794509 < 0$	Case I ($d = 0$)
5	0.3695305796	240.4046723514	240.4397623868	+8.26104550	+105.7081648	$-873.2599586 < 0$	Case I ($d = 0$)

Part B: Codimension-Three Interior Saddle Overdetermination Tests ($\partial_d I_{\mathrm{def}} = \partial_t I_{\mathrm{def}} = 0 \implies I_{\mathrm{def}}(d_c, t_c) \neq 0$)

(γ_0, d_0)	d_c	t_c	$\lvert\nabla I_{\mathrm{def}}(d_c, t_c)\rvert$	$S_c I_{\mathrm{def}}(d_c, t_c)$	$S_0^2 c_1^{\mathrm{def}}(\gamma_0)$	Saddle Locus	Status
$(14.6, 0.35)$	0.0875902564	14.4497284769	4.37×10^{-64}	$+0.1303705131 \neq 0$	$-378.554601 < 0$	$I_{\mathrm{def}} \neq 0$	Codim-3 block
$(20.4, 0.35)$	0.1174426402	20.6035255099	3.28×10^{-63}	$-0.8333087817 \neq 0$	$-4994.903951 < 0$	$I_{\mathrm{def}} \neq 0$	Codim-3 block
$(21.5, 0.35)$	0.0532790009	21.3477026052	5.95×10^{-64}	$-0.0790228758 \neq 0$	$-2781.827623 < 0$	$I_{\mathrm{def}} \neq 0$	Codim-3 block
$(25.5, 0.35)$	0.0622885100	25.3424252306	7.79×10^{-64}	$+0.2186756407 \neq 0$	$-9290.443548 < 0$	$I_{\mathrm{def}} \neq 0$	Codim-3 block
$(30.8, 0.28)$	0.0491031765	30.6812175656	3.55×10^{-64}	$-0.1278773776 \neq 0$	$-4370.880169 < 0$	$I_{\mathrm{def}} \neq 0$	Codim-3 block

Part C: First 10 Chladni Nodal Walls $\Gamma_1, \dots, \Gamma_{10}$ of Riemann's $\Xi(z)$ ($S_k = \mathrm{e}^{\pi\tau_k(0)/4}$)

Wall	$\tau_k(0)$	$S_k\Xi(\tau_k(0))$	$S_k\Xi''(\tau_k(0))$	$S_k^2 c_1(\tau_k(0))$	$\lvert R(0.25)\rvert/\lvert\Xi\rvert$	$\lvert R(0.50)\rvert/\lvert\Xi\rvert$	Wall Trend
Γ_1	15.5857085898	−166.008602	+92.981069	$+1.543566 \times 10^4 > 0$	$1.017538 > 1$	$1.072108 > 1$	Monotone ↑
Γ_2	22.0979772804	+267.434544	−283.061825	$+7.570051 \times 10^4 > 0$	$1.033290 > 1$	$1.138825 > 1$	Monotone ↑
Γ_3	26.2722473569	−612.407187	+489.454784	$+2.997456 \times 10^5 > 0$	$1.025120 > 1$	$1.103855 > 1$	Monotone ↑
Γ_4	31.2317958710	+364.583681	−728.500528	$+2.655994 \times 10^5 > 0$	$1.063195 > 1$	$1.267862 > 1$	Monotone ↑
Γ_5	34.1933102690	−1066.490578	+944.080885	$+1.006853 \times 10^6 > 0$	$1.027808 > 1$	$1.115138 > 1$	Monotone ↑
Γ_6	38.4982407638	+960.782380	−1471.658433	$+1.413943 \times 10^6 > 0$	$1.048801 > 1$	$1.204611 > 1$	Monotone ↑
Γ_7	41.7367295224	−741.738637	+1531.083116	$+1.135664 \times 10^6 > 0$	$1.064974 > 1$	$1.275482 > 1$	Monotone ↑
Γ_8	44.5417036074	+2173.336660	−2115.524567	$+4.597747 \times 10^6 > 0$	$1.030676 > 1$	$1.127137 > 1$	Monotone ↑
Γ_9	48.6225326853	−654.372987	+2336.966911	$+1.529248 \times 10^6 > 0$	$1.114084 > 1$	$1.493789 > 1$	Monotone ↑
Γ_{10}	50.8390048228	+1751.646068	−2391.982389	$+4.189907 \times 10^6 > 0$	$1.043069 > 1$	$1.179677 > 1$	Monotone ↑

3. Exact 50%–50% Green–Dirichlet–Neumann Chamber Equipartition and Wronskian–Slope Reciprocity

Having established the macro-scale boundary anatomy of the Chladni walls Γ_k and Γ_{k+1}, we now turn to the meso-scale interior of the Chladni portal frame

$$\Omega_k(d_0) := \Big\{(d,t) \in \mathbb{R}^2 : 0 < d < d_0,\ \ \tau_k(d) < t < \tau_{k+1}(d)\Big\}, \qquad d_0 > 0, \tag{3.1}$$

bounded on the left by the critical-line segment $\partial\Omega_{k,\mathrm{L}} = \{0\} \times [\tau_k(0), \tau_{k+1}(0)]$, on the bottom and top by the curved Chladni walls $\Gamma_k(d_0) = \{(d, \tau_k(d)) : 0 \le d \le d_0\}$ and $\Gamma_{k+1}(d_0) = \{(d, \tau_{k+1}(d)) : 0 \le d \le d_0\}$ where $I(d,t) = 0$, and on the right by the vertical cutoff $\partial\Omega_{k,\mathrm{R}}(d_0) = \{d_0\} \times [\tau_k(d_0), \tau_{k+1}(d_0)]$.

Remark 3.1 (Wronskian Normalization Dictionary). Along any vertical line $d = d_0 > 0$, the parity-reduced transverse Wronskian density of (R, I) is given by $W(d_0, t) = 2\big(R(d_0,t)\,\partial_t I(d_0,t) - I(d_0,t)\,\partial_t R(d_0,t)\big) = 2\big(R(d_0,t)\,\partial_d R(d_0,t) + I(d_0,t)\,\partial_d I(d_0,t)\big) = 2d_0\,\widetilde{W}(d_0,t)$. Integrating $W(d_0,t)$ over $t \in [\tau_k(d_0), \tau_{k+1}(d_0)]$ and applying Green's identity over $\Omega_k(d_0)$ yields the classical chamber flux identity [17, 22]

$$\mathcal{W}_{\Omega_k}(d_0) := \int_{\tau_k(d_0)}^{\tau_{k+1}(d_0)} \big[R(d_0,t)\,\partial_d R(d_0,t) + I(d_0,t)\,\partial_d I(d_0,t)\big]\,\mathrm{d}t = 2\iint_{\Omega_k(d_0)} \big|\Xi'(t-\mathrm{i}d)\big|^2\,\mathrm{d}d\,\mathrm{d}t > 0. \tag{3.2}$$

What was not previously recognized is that the two boundary terms on the left-hand side of (3.2)—the purely real-part flux $\int R\,\partial_d R\,\mathrm{d}t$ and the imaginary cross-term flux $\int I\,\partial_d I\,\mathrm{d}t$—are not merely positive numerical contributors to their sum, but are *identically equal* at every cutoff depth $d_0 > 0$, even though the bounding walls $\tau_k(d)$ and $\tau_{k+1}(d)$ curve non-trivially with d.

Theorem 3.2 (Exact 50%–50% Green–Dirichlet–Neumann Chamber Equipartition)**.** *Let $F(z)$ be any non-constant real-entire function with $F(t-\mathrm{i}d) = R(d,t) + \mathrm{i}I(d,t)$, and let $\Omega_k(d_0)$ be any Chladni chamber (3.1) bounded by smooth Chladni walls $\Gamma_k, \Gamma_{k+1} \subset \{I(d,t) = 0\}$. Then for every $d_0 > 0$, the real boundary flux $\mathcal{R}_{\Omega_k}(d_0)$ and the imaginary boundary flux $\mathcal{I}_{\Omega_k}(d_0)$ across the vertical right-hand face $d = d_0$ are identically equal to one another and to the 2D Dirichlet energy $\mathcal{E}_{\Omega_k}(d_0)$:*

$$\boxed{\int_{\tau_k(d_0)}^{\tau_{k+1}(d_0)} R\,\partial_d R\Big|_{d=d_0}\,\mathrm{d}t \equiv \int_{\tau_k(d_0)}^{\tau_{k+1}(d_0)} I\,\partial_d I\Big|_{d=d_0}\,\mathrm{d}t \equiv \iint_{\Omega_k(d_0)} \big|F'(t-\mathrm{i}d)\big|^2\,\mathrm{d}d\,\mathrm{d}t = \frac{1}{2}\mathcal{W}_{\Omega_k}(d_0) > 0.} \tag{3.3}$$

Equivalently, $\iint_{\Omega_k(d_0)} \Delta(R^2 - I^2)\,\mathrm{d}d\,\mathrm{d}t = 0$, and the chamber boundary $\partial\Omega_k(d_0) \setminus \partial\Omega_{k,\mathrm{R}}(d_0)$ is simultaneously a homogeneous Dirichlet boundary ($I = 0$) for the harmonic field I and a homogeneous Neumann boundary ($\partial_n R = 0$) for the conjugate harmonic field R.

Proof. Since $F(t-\mathrm{i}d) = R(d,t) + \mathrm{i}I(d,t)$ is holomorphic in $z = t - \mathrm{i}d$, both R and I are harmonic on $\mathbb{R}^2$ ($\Delta R = \partial_d^2 R + \partial_t^2 R = 0$ and $\Delta I = \partial_d^2 I + \partial_t^2 I = 0$), and their Euclidean gradients satisfy

$$|\nabla R|^2 = (\partial_d R)^2 + (\partial_t R)^2 = (\partial_t I)^2 + (-\partial_d I)^2 = |\nabla I|^2 = \big|F'(t-\mathrm{i}d)\big|^2.$$

We apply Green's first identity $\iint_{\Omega_k(d_0)} (u\,\Delta u + |\nabla u|^2)\,\mathrm{d}d\,\mathrm{d}t = \oint_{\partial\Omega_k(d_0)} u\,\partial_n u\,\mathrm{d}s$ separately to $u = I$ and $u = R$, where $\mathbf{n} = (n_d, n_t)$ denotes the outward unit normal along the four-part boundary $\partial\Omega_k(d_0) = \partial\Omega_{k,\mathrm{L}} \cup \Gamma_k(d_0) \cup \Gamma_{k+1}(d_0) \cup \partial\Omega_{k,\mathrm{R}}(d_0)$:

(i) **Dirichlet Boundary Evaluation for $u = I$:** On the left floor $\partial\Omega_{k,\mathrm{L}} = \{d = 0\}$ and on both curved Chladni walls $\Gamma_k(d_0)$ and $\Gamma_{k+1}(d_0)$, the field $I(d,t)$ vanishes identically by definition ($I = 0$). On the vertical right boundary $\partial\Omega_{k,\mathrm{R}}(d_0) = \{d = d_0\}$, the outward unit normal is $\mathbf{n} = (+1, 0)$, so $\partial_n I = \partial_d I$ and $\mathrm{d}s = \mathrm{d}t$. Green's first identity for $u = I$ with $\Delta I = 0$ therefore collapses to

$$\iint_{\Omega_k(d_0)} \big|F'(t-\mathrm{i}d)\big|^2\,\mathrm{d}d\,\mathrm{d}t = \iint_{\Omega_k(d_0)} |\nabla I|^2\,\mathrm{d}d\,\mathrm{d}t = \int_{\tau_k(d_0)}^{\tau_{k+1}(d_0)} I(d_0,t)\,\partial_d I(d_0,t)\,\mathrm{d}t.$$

(ii) **Neumann Boundary Evaluation for $u = R$:** On the left floor $\partial\Omega_{k,\mathrm{L}} = \{d = 0\}$, the outward unit normal is $\mathbf{n} = (-1, 0)$ and $\partial_n R(0,t) = -\partial_d R(0,t) = -\partial_t I(0,t) = 0$ by even reflection symmetry $R(-d,t) = R(d,t)$. Along the curved bottom and top Chladni walls $\Gamma_k(d_0)$ and $\Gamma_{k+1}(d_0)$, let $\mathbf{s}$ be the counterclockwise unit tangent vector; by the Cauchy–Riemann equations in intrinsic coordinates, the outward normal derivative of R is equal to the tangential derivative of I:

$$\partial_n R = \nabla R \cdot \mathbf{n} = \partial_d R\,n_d + \partial_t R\,n_t = \partial_t I\,n_d - \partial_d I\,n_t = \nabla I \cdot \mathbf{s} = \frac{\mathrm{d}I}{\mathrm{d}s}\Big|_{\Gamma_k, \Gamma_{k+1}} \equiv 0,$$

because $I \equiv 0$ identically along both curved walls $\Gamma_k(d_0)$ and $\Gamma_{k+1}(d_0)$ regardless of their slope $\tau_k'(d)$. Thus $R\,\partial_n R \equiv 0$ on $\partial\Omega_{k,\mathrm{L}} \cup \Gamma_k(d_0) \cup \Gamma_{k+1}(d_0)$. Only the vertical right boundary $\partial\Omega_{k,\mathrm{R}}(d_0)$ ($\mathbf{n} = (+1, 0)$, $\partial_n R = \partial_d R$) contributes to Green's first identity for $u = R$, giving

$$\iint_{\Omega_k(d_0)} \left|F'(t - \mathrm{i}d)\right|^2 \mathrm{d}d\,\mathrm{d}t = \iint_{\Omega_k(d_0)} |\nabla R|^2 \,\mathrm{d}d\,\mathrm{d}t = \int_{\tau_k(d_0)}^{\tau_{k+1}(d_0)} R(d_0, t)\,\partial_d R(d_0, t)\,\mathrm{d}t.$$

Equating the two right-hand expressions establishes the boxed three-way identity (3.3). Strict positivity holds for every $d_0 > 0$ because $\Omega_k(d_0)$ has positive planar measure and the zeros of F' are isolated. □

Next, we establish the pointwise meso-scale differential identities coupling the geometric bending slope $\tau_k'(d)$ of a Chladni wall (and the dual slope $t_R'(d)$ of any real nodal curve $R = 0$) to the reduced transverse Wronskian $\widetilde{W}(d, t)$.

Theorem 3.3 (Wronskian–Slope Reciprocity and Wall-Amplitude Evolution)**.** *Let $F(t - \mathrm{i}d) = R(d, t) + \mathrm{i}I(d, t)$ have reduced transverse Wronskian $\widetilde{W}(d, t) = (R\,\partial_t I - I\,\partial_t R)/d = (R\,\partial_d R + I\,\partial_d I)/d$.*

(a) ***Chladni Wall Wronskian–Slope Reciprocity:*** *Along any smooth segment of a Chladni wall $\Gamma_k = \{(d, \tau_k(d)) : d > 0\}$ where $I(d, \tau_k(d)) = 0$ and $\partial_t I(d, \tau_k(d)) \neq 0$, the reduced Wronskian and the wall slope $\tau_k'(d)$ satisfy the exact identities*

$$d\,\widetilde{W}\big(d, \tau_k(d)\big) \equiv R\,\partial_t I\Big|_{\Gamma_k} \equiv R\,\partial_d R\Big|_{\Gamma_k}, \qquad \tau_k'(d) \equiv \frac{R(d, \tau_k(d))\,\partial_t R(d, \tau_k(d))}{d\,\widetilde{W}\big(d, \tau_k(d)\big)}, \tag{3.4}$$

together with the arc-length Jacobian identity and the Wall-Amplitude Evolution Law:

$$1 + \tau_k'(d)^2 \equiv \frac{R(d, \tau_k(d))^2\left|F'(\tau_k(d) - \mathrm{i}d)\right|^2}{d^2\,\widetilde{W}\big(d, \tau_k(d)\big)^2}, \qquad \frac{\mathrm{d}}{\mathrm{d}d}\Big[\frac{1}{2}R\big(d, \tau_k(d)\big)^2\Big] \equiv d\,\widetilde{W}\big(d, \tau_k(d)\big)\,\big(1 + \tau_k'(d)^2\big). \tag{3.5}$$

(b) ***Dual Real-Curve Reciprocity and Fold-Vertex Wronskian Vanishing:*** *Along any smooth segment of a real nodal contour $\Gamma_R = \{(d, t_R(d)) : d > 0\}$ where $R(d, t_R(d)) = 0$, the dual identities hold:*

$$t_R'(d) \equiv \frac{I(d, t_R(d))\,\partial_t I(d, t_R(d))}{d\,\widetilde{W}\big(d, t_R(d)\big)}, \qquad \frac{\mathrm{d}}{\mathrm{d}d}\Big[\frac{1}{2}I\big(d, t_R(d)\big)^2\Big] \equiv d\,\widetilde{W}\big(d, t_R(d)\big)\,\big(1 + t_R'(d)^2\big). \tag{3.6}$$

Consequently, if a real nodal contour $R(d, t) = 0$ forms a detached parabolic fold in the interior $d > 0$ with leftmost turning-point vertex $(d_{\mathrm{tip}}, t_{\mathrm{tip}})$ where $\partial_t R(d_{\mathrm{tip}}, t_{\mathrm{tip}}) = 0$ and $|t_R'(d_{\mathrm{tip}})| = \infty$ (as occurs in Davenport–Heilbronn Ξ_{DH}), the reduced transverse Wronskian must vanish identically at the fold tip: $\widetilde{W}(d_{\mathrm{tip}}, t_{\mathrm{tip}}) = 0$.

Proof. (a) From the definition of the parity-reduced transverse Wronskian [17, 22],

$$d\,\widetilde{W}(d, t) := R(d, t)\,\partial_t I(d, t) - I(d, t)\,\partial_t R(d, t) = R(d, t)\,\partial_d R(d, t) + I(d, t)\,\partial_d I(d, t). \tag{3.7}$$

Expanding $R(d, t) = F(t) - \frac{1}{2}F''(t)d^2 + O(d^4)$ and $I(d, t) = -F'(t)d + \frac{1}{6}F^{(3)}(t)d^3 + O(d^5)$ as $d \to 0^+$ confirms that $\widetilde{W}(d, t) = (F'(t))^2 - F(t)F''(t) + O(d^2) = c_1(t) + O(d^2)$. Evaluating (3.7) along a Chladni wall Γ_k where $I(d, \tau_k(d)) \equiv 0$ immediately yields the first identity in (3.4): $d\,\widetilde{W}(d, \tau_k(d)) \equiv R(d, \tau_k(d))\,\partial_t I(d, \tau_k(d)) \equiv R(d, \tau_k(d))\,\partial_d R(d, \tau_k(d))$. Differentiating $I(d, \tau_k(d)) \equiv 0$ implicitly with respect to d gives $\partial_d I + \partial_t I\,\tau_k'(d) = 0$, so by the Cauchy–Riemann equation $\partial_t R = -\partial_d I$,

$$\tau_k'(d) = -\frac{\partial_d I}{\partial_t I} = \frac{\partial_t R}{\partial_t I} = \frac{R(d, \tau_k(d))\,\partial_t R(d, \tau_k(d))}{R(d, \tau_k(d))\,\partial_t I(d, \tau_k(d))} = \frac{R(d, \tau_k(d))\,\partial_t R(d, \tau_k(d))}{d\,\widetilde{W}\big(d, \tau_k(d)\big)},$$

which is the second identity in (3.4). Squaring and adding 1 gives

$$1+\tau_k'(d)^2=\frac{R^2\big((\partial_t I)^2+(\partial_t R)^2\big)}{d^2\,\widetilde{W}^2}=\frac{R(d,\tau_k(d))^2\,\big|F'(\tau_k(d)-\mathrm{i}d)\big|^2}{d^2\,\widetilde{W}\big(d,\tau_k(d)\big)^2},$$

proving the first identity in (3.5). Finally, differentiating $\frac{1}{2}R(d,\tau_k(d))^2$ with respect to d along Γ_k gives

$$\frac{\mathrm{d}}{\mathrm{d}d}\Big[\frac{1}{2}R\big(d,\tau_k(d)\big)^2\Big]=R\big(\partial_d R+\partial_t R\,\tau_k'(d)\big)=R\,\partial_d R\,\Big(1+\frac{\partial_t R}{\partial_d R}\tau_k'(d)\Big)=d\,\widetilde{W}\big(d,\tau_k(d)\big)\,\big(1+\tau_k'(d)^2\big),$$

establishing the Wall-Amplitude Evolution Law in (3.5).

(b) Evaluating (3.7) along a real nodal curve Γ_R where $R(d,t_R(d))=0$ gives $d\,\widetilde{W}(d,t_R(d))\equiv -I(d,t_R(d))\,\partial_t R(d,t_R(d))\equiv I(d,t_R(d))\,\partial_d I(d,t_R(d))$. Implicit differentiation of $R(d,t_R(d))\equiv 0$ gives $t_R'(d)=-\partial_d R/\partial_t R=\partial_t I/\partial_d I=I\,\partial_t I/(d\,\widetilde{W})$, proving the first identity in (3.6). Differentiating $\frac{1}{2}I(d,t_R(d))^2$ with respect to d gives $I(\partial_d I+\partial_t I\,t_R'(d))=I\,\partial_d I\,(1+t_R'(d)^2)=d\,\widetilde{W}\,(1+t_R'(d)^2)$, proving the second identity in (3.6). At a detached parabolic fold vertex $(d_{\mathrm{tip}},t_{\mathrm{tip}})$ where $R=0$ and the tangent to $R=0$ is vertical ($\partial_t R(d_{\mathrm{tip}},t_{\mathrm{tip}})=0$, $|t_R'(d_{\mathrm{tip}})|=\infty$), equation (3.7) gives $d_{\mathrm{tip}}\widetilde{W}(d_{\mathrm{tip}},t_{\mathrm{tip}})=0\cdot\partial_t I-I\cdot 0=0$. □

TABLE 2. Fifty-two-decimal meso-scale certification of Theorem 3.2 (50%–50% Green–Dirichlet–Neumann chamber equipartition) and Theorem 3.3 (Wronskian–slope reciprocity): (Part A) the first three Chladni chambers $\Omega_1,\Omega_2,\Omega_3$ of Riemann's $\Xi(z)$ across cutoff depths $d_0\in\{0.05,0.15,0.25,0.50\}$; (Part B) head-to-head comparison at $t\in[83.89,86.63]$ between the Euler-product chamber $\Omega_{\chi_5}(d_0)$ of $\Xi_{\chi_5}(z)$ and the split counterexample chambers $\Omega_{\mathrm{DH,L}}(d_0)$, $\Omega_{\mathrm{DH,R}}(d_0)$ of Davenport–Heilbronn $\Xi_{\mathrm{DH}}(z)$ flanking the false wall $\Gamma_{\mathrm{false},1}$ ($\tau_{\mathrm{false},1}(0)=85.74720083$). At $d_0=d_{*,1}=0.30851718$, $\widetilde{W}_{\mathrm{DH}}$ on $\Gamma_{\mathrm{false},1}$ crosses zero ($|\widetilde{W}_{\mathrm{DH}}|=3.01\times10^{-144}$), and at the detached real parabola fold vertex $(d_{\mathrm{tip}},t_{\mathrm{tip}})=(0.29809801,85.62549444)$, $|\widetilde{W}_{\mathrm{DH}}|=2.47\times10^{-144}=0$ with $|t_R'(d_{\mathrm{tip}})|=\infty$.

Chamber	d_0	$\mathcal{I}_\Omega(d_0)=\mathcal{R}_\Omega(d_0)=\mathcal{E}_\Omega(d_0)$	$\frac{\lvert\mathcal{I}_\Omega-\mathcal{R}_\Omega\rvert}{\mathcal{E}_\Omega}$	$\frac{\lvert\mathcal{I}_\Omega-\mathcal{E}_\Omega\rvert}{\mathcal{E}_\Omega}$	Wall $\widetilde{W}(d_0,\tau_{k+1})$	Slope $\tau_{k+1}'(d_0)$	Recip. Err
Part A: Chladni Chambers $\Omega_1,\Omega_2,\Omega_3$ of Riemann's $\Xi(z)$ ($t\in[15.58,31.24]$)							
Ω_1	0.050000	$7.77993878916345\times10^{-9}$	5.97×10^{-58}	2.14×10^{-61}	$+6.38062292\times10^{-11}$	-0.0239705250	$<10^{-64}$
Ω_1	0.150000	$2.35181517512051\times10^{-8}$	5.95×10^{-58}	1.91×10^{-60}	$+6.46846488\times10^{-11}$	-0.0719209979	$<10^{-64}$
Ω_1	0.250000	$3.97977470595614\times10^{-8}$	5.93×10^{-58}	5.25×10^{-60}	$+6.64724927\times10^{-11}$	-0.1198983470	$<10^{-64}$
Ω_1	0.500000	$8.54612143883906\times10^{-8}$	5.79×10^{-58}	1.99×10^{-59}	$+7.54282251\times10^{-11}$	-0.2400285130	$<10^{-64}$
Ω_2	0.050000	$1.18229802986119\times10^{-12}$	4.39×10^{-62}	2.19×10^{-65}	$+3.58697788\times10^{-13}$	-0.0203017125	$<10^{-64}$
Ω_2	0.250000	$6.15549397688721\times10^{-12}$	4.27×10^{-62}	1.33×10^{-64}	$+3.70614547\times10^{-13}$	-0.1014100990	$<10^{-64}$
Ω_2	0.500000	$1.39509999297565\times10^{-11}$	3.92×10^{-62}	5.06×10^{-61}	$+4.10198731\times10^{-13}$	-0.2021606880	$<10^{-64}$
Ω_3	0.050000	$6.87146355336909\times10^{-15}$	8.56×10^{-60}	7.60×10^{-64}	$+1.31707625\times10^{-16}$	-0.0283569314	$<10^{-64}$
Ω_3	0.250000	$3.55273128496507\times10^{-14}$	8.16×10^{-60}	1.72×10^{-62}	$+1.41764534\times10^{-16}$	-0.1416485060	$<10^{-64}$
Ω_3	0.500000	$7.88333180096506\times10^{-14}$	7.01×10^{-60}	3.66×10^{-61}	$+1.76842660\times10^{-16}$	-0.2823150930	$<10^{-64}$
Part B: Head-to-Head Comparison of Euler-Product $\Xi_{\chi_5}(z)$ vs. Davenport–Heilbronn $\Xi_{\mathrm{DH}}(z)$ ($t\in[83.89,86.63]$)							
Ω_{χ_5}	0.050000	$5.82340286508510\times10^{-56}$	8.99×10^{-59}	3.71×10^{-62}	$+1.46410672\times10^{-56}$	-0.0507833251	$<10^{-64}$
Ω_{χ_5}	0.250000	$3.26670201080539\times10^{-55}$	6.48×10^{-59}	1.44×10^{-60}	$+2.09832879\times10^{-56}$	-0.2462769960	$<10^{-64}$
Ω_{χ_5}	0.308517	$4.28746718801361\times10^{-55}$	5.40×10^{-59}	2.32×10^{-60}	$+2.50713066\times10^{-56}$	-0.2988429150	$<10^{-64}$
Ω_{χ_5}	0.500000	$9.21590394706157\times10^{-55}$	2.24×10^{-59}	5.18×10^{-53}	$+5.18204517\times10^{-56}$	-0.4465968730	$<10^{-64}$
$\Omega_{\mathrm{DH,L}}$	0.050000	$2.19680583282156\times10^{-56}$	2.85×10^{-61}	5.47×10^{-64}	$-6.34616468\times10^{-57}$	-0.0522568475	$<10^{-64}$
$\Omega_{\mathrm{DH,L}}$	0.250000	$1.27197067309754\times10^{-55}$	1.89×10^{-61}	4.16×10^{-61}	$-2.61224498\times10^{-57}$	-0.2480786530	$<10^{-64}$
$\Omega_{\mathrm{DH,L}}$	0.308517	$1.69699405094649\times10^{-55}$	1.52×10^{-61}	3.17×10^{-61}	$+3.01174152\times10^{-144}$	-0.2978556240	$<10^{-64}$
$\Omega_{\mathrm{DH,L}}$	0.500000	$3.91370158628202\times10^{-55}$	5.89×10^{-62}	1.65×10^{-52}	$+1.91178220\times10^{-56}$	-0.4267401100	$<10^{-64}$
$\Omega_{\mathrm{DH,R}}$	0.308517	$1.03437862265482\times10^{-57}$	3.60×10^{-69}	9.97×10^{-59}	$+1.14345722\times10^{-56}$	-0.0229552243	$<10^{-64}$

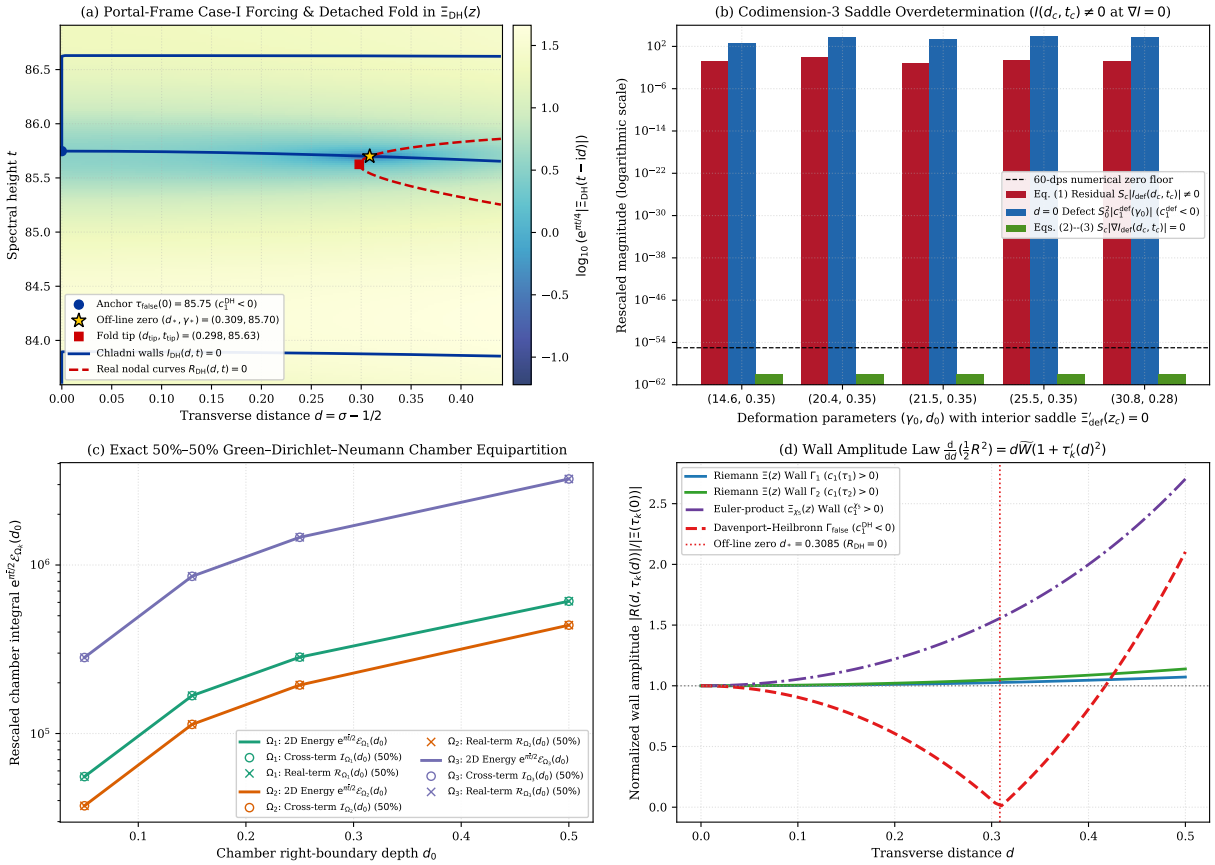


FIGURE 1. Macro- and meso-scale Chladni portal-frame synthesis (Sections 2–3): (a) Davenport–Heilbronn counterexample chamber $t \in [83.6, 86.9]$, illustrating the Case-I false wall $\Gamma_{\mathrm{false},1}$ originating at $\tau_{\mathrm{false},1}(0) = 85.75$ ($c_1^{\mathrm{DH}} < 0$), the off-critical zero $(d_*, \gamma_*) = (0.3085, 85.70)$ where $\Gamma_{\mathrm{false},1}$ crosses the detached real nodal parabola $R_{\mathrm{DH}} = 0$, and the fold vertex $(d_{\mathrm{tip}}, t_{\mathrm{tip}}) = (0.2981, 85.63)$ where $\widetilde{W}_{\mathrm{DH}} = 0$ and $|t'_R(d_{\mathrm{tip}})| = \infty$; (b) codimension-three saddle overdetermination tests across five deformed saddles $\Xi'_{\mathrm{def}}(z_c) = 0$ in $d_c > 0$, showing that $I_{\mathrm{def}}(d_c, t_c) \neq 0$ by 60+ orders of magnitude above the numerical floor; (c) exact 50%–50% Green–Dirichlet–Neumann chamber equipartition $\mathcal{I}_{\Omega_k}(d_0) \equiv \mathcal{R}_{\Omega_k}(d_0) \equiv \mathcal{E}_{\Omega_k}(d_0)$ across $\Omega_1, \Omega_2, \Omega_3$ of $\Xi(z)$; (d) Wall-Amplitude Evolution Law (3.5), comparing strictly increasing Euler-product walls ($\Xi(z)$ and $\Xi_{\chi_5}(z)$) against the Case-I wall collapse of $\Xi_{\mathrm{DH}}(z)$.

4. Attenuated Semiprime Spring, Euler Dilogarithm $\mathrm{Li}_2(2\theta)$ Horizon Damping, and Dirichlet Hyperbola Reflection Duality

Sections 2 and 3 demonstrated that the non-vanishing of $R(d, \tau_k(d))$ along every Chladni wall Γ_k and the monotonic growth of the chamber energy $\mathcal{E}_{\Omega_k}(d_0)$ are governed by the positivity of the reduced transverse Wronskian $\widetilde{W}(d, t)$ and its critical-line limit $c_1(t)$. We now descend to the micro-scale to resolve the third bottleneck stated in Section 1: how the finite Euler-product prime gauge $\mathcal{P}_{\leq X}(s)$ and the Riemann–Siegel stationary-phase horizon $N_\star(t) = \sqrt{t/(2\pi)}$ interact across the quadratic semiprime window $n = pq \in (X^2, N_\star^2]$ when the prime tail $\mathcal{P}_{>X}(s)$ enters an anti-phase configuration $U = \mathrm{Re}\, \mathcal{P}_{>X}(s) < 0$.

4.1. **Pre-move holomorphic and Stieltjes-integral justification.** Before executing any term-by-term differentiation, Dirichlet hyperbola splitting, or Euler–Maclaurin integral replacement on the finite prime-gauge decomposition, we record the exact domains of holomorphicity, absolute convergence, and Stieltjes-integral regularity that justify each analytical operation:

(1) **Entire and Polynomial Holomorphicity:** For every finite sieve cutoff $X \geq 2$ and every finite Dirichlet truncation $N = \lfloor N_\star(t) \rfloor$, the prime-gauge Dirichlet polynomial $\mathcal{P}_{>X}(s) := \sum_{X<p\leq N} p^{-s}$ and its quadratic semiprime square

$$\frac{1}{2}\mathcal{P}_{>X}(s)^2 = \frac{1}{2}\sum_{X<p\leq N}\sum_{X<q\leq N}(pq)^{-s} = \mathcal{T}_{\leq N}(s) + \mathcal{C}_{>N}(s) \tag{4.1}$$

are finite exponential sums, hence entire functions of $s = \frac{1}{2} + d + \mathrm{i}t \in \mathbb{C}$ of exponential type $\leq 2\ln N$. Consequently, the Cauchy–Riemann differential operators ∂_d and ∂_t commute unconditionally with the finite sums defining $\mathcal{T}_{\leq N}(s)$ and $\mathcal{C}_{>N}(s)$ for all $(d,t) \in \mathbb{R}^2$.

(2) **Riemann–Stieltjes Representation and Abel Summation:** Let $\vartheta(u) := \sum_{p\leq u} \ln p$ denote Chebyshev's prime-counting step function, which is right-continuous and of locally bounded variation on $[2,\infty)$. For any continuously differentiable weight $g \in C^1([X, N_\star])$, the discrete prime sum $\sum_{X<p\leq N_\star} \frac{g(p)}{p}$ is represented exactly as the Riemann–Stieltjes integral

$$\sum_{X<p\leq N_\star}\frac{g(p)}{p} = \int_X^{N_\star}\frac{g(u)}{u\ln u}\,\mathrm{d}\vartheta(u) = \int_\theta^{1/2}\frac{g(\mathrm{e}^{\alpha L})}{\alpha}\,\mathrm{d}\alpha + \mathcal{R}_{\mathrm{PNT}}(X, N_\star; g), \tag{4.2}$$

where $L = \ln N_\star(t)$, $\theta = (\ln X)/L \in (0, 1/4)$, and integration by parts with the Prime Number Theorem remainder $\vartheta(u) - u = O(u\exp(-c\sqrt{\ln u}))$ [24, 30] bounds the discretization remainder by $\mathcal{R}_{\mathrm{PNT}}(X, N_\star; g) = O(\|g\|_{W^{1,\infty}}(\ln X)^{-1})$.

(3) **Uniform Convergence of the Dilogarithm Kernel on $[\theta, 1/2]^2$:** Under the logarithmic change of variables $\alpha_1 = (\ln p)/L$ and $\alpha_2 = (\ln q)/L$, the normalized horizon weight $(\omega_p(t)/L)^p = (1-2\alpha_1)^p$ is a smooth polynomial on the compact square $[\theta, 1/2]^2$ for every $\theta \in (0, 1/4)$. Because $\theta > 0$ is bounded away from the logarithmic singularity at $\alpha_1\alpha_2 = 0$, the two-dimensional density $(1-2\alpha_1)^p(1-2\alpha_2)^p/(\alpha_1\alpha_2)$ is uniformly bounded and real-analytic on $[\theta, 1/2]^2$, allowing Fubini's theorem and power-series integration of $\ln(1-u)/u$ into Euler's dilogarithm $\mathrm{Li}_2(u)$ to be applied without improper-integral regularizations.

4.2. **The Attenuated Quadratic Semiprime Spring and Discriminant Reversal.** In Paper XIII [22], we showed that when the Euler product $\zeta(s) = \prod_{p\leq X}(1-p^{-s})^{-1}\prod_{p>X}(1-p^{-s})^{-1}$ is factored at the cubic-window threshold $X = N_\star(t)^{1/3}$ ($\theta = 1/6$), all primes $p > X$ satisfy $p^3 > N_\star(t)$, so the residual factor $\exp(\mathcal{P}_{>X}(s) + O(p^{-2s}))$ inside the forward Riemann–Siegel sum $n \leq N_\star(t)$ truncates at quadratic semiprime order $n = pq$:

$$1 + \mathcal{P}_{>X}(s) + \mathcal{T}_{\leq N_\star}(s), \qquad \mathcal{T}_{\leq N_\star}(s) := \frac{1}{2}\sum_{\substack{X<p\leq N_\star \\ pq\leq N_\star(t)}}\sum_{X<q\leq N_\star}(pq)^{-s}, \tag{4.3}$$

while the complementary reflected-corner pairs

$$\mathcal{C}_{>N_\star}(s) := \frac{1}{2}\sum_{\substack{X<p\leq N_\star \\ pq>N_\star(t)}}\sum_{X<q\leq N_\star}(pq)^{-s} = \frac{1}{2}\mathcal{P}_{>X}(s)^2 - \mathcal{T}_{\leq N_\star}(s) \tag{4.4}$$

exceed the Riemann–Siegel horizon $pq > N_\star(t)$ and are transferred into the conjugate reflected branch of the approximate functional equation. Writing $\mathcal{P}_{>X}(s) = U + \mathrm{i}V$ with $U = \mathrm{Re}\,\mathcal{P}_{>X}(s)$ and $V = \mathrm{Im}\,\mathcal{P}_{>X}(s)$, suppose that the forward triangle $\mathcal{T}_{\leq N_\star}(s)$ captures an effective fraction $\eta \in (0,1]$ of the full quadratic square $\frac{1}{2}\mathcal{P}_{>X}(s)^2$. Our first micro-scale theorem proves that whenever $\eta > 1/2$,

the quadratic semiprime spring possesses a strictly negative discriminant $\Delta(\eta) = 1 - 2\eta < 0$ and a strictly positive global energy floor along the anti-phase axis $V = 0$.

Theorem 4.1 (Attenuated Quadratic Semiprime Spring and Discriminant Reversal)**.** *For any capture fraction $\eta \in (0, 1]$ and any complex prime-tail amplitude $w = U + \mathrm{i}V \in \mathbb{C}$, define the attenuated quadratic semiprime polynomial*

$$\mathcal{Q}_\eta(U + \mathrm{i}V) := 1 + (U + \mathrm{i}V) + \frac{\eta}{2}(U + \mathrm{i}V)^2 = \Big(1 + U + \frac{\eta}{2}(U^2 - V^2)\Big) + \mathrm{i}V(1 + \eta U). \tag{4.5}$$

Then:

(a) ***Exact Anti-Phase Discriminant and Global Floor*** *($\eta > 1/2$)**:** Along the worst-case anti-phase real axis $V = 0$, the real quadratic polynomial $\mathcal{Q}_\eta(U) = 1 + U + \frac{\eta}{2}U^2$ has algebraic discriminant*

$$\Delta(\eta) := 1^2 - 4 \cdot \frac{\eta}{2} \cdot 1 = 1 - 2\eta. \tag{4.6}$$

Whenever $\eta > 1/2$, $\Delta(\eta) < 0$, so $\mathcal{Q}_\eta(U)$ has no real roots and satisfies the sharp global lower bound

$$\big|\mathcal{Q}_\eta(U)\big|^2 = \Big|1 + U + \frac{\eta}{2}U^2\Big|^2 = \Big[1 - \frac{1}{2\eta} + \frac{\eta}{2}\Big(U + \frac{1}{\eta}\Big)^2\Big]^2 \geq \Big(1 - \frac{1}{2\eta}\Big)^2 > 0 \qquad (\forall\, U \in \mathbb{R}), \tag{4.7}$$

attaining its unique global minimum at $U_{\min}(\eta) = -1/\eta < -1$. Conversely, if $\eta \leq 1/2$, then $\Delta(\eta) \geq 0$ and $\mathcal{Q}_\eta(U)$ vanishes on the real anti-phase axis at $U_\pm(\eta) = (-1 \pm \sqrt{1 - 2\eta})/\eta \leq -2$.

(b) ***Full Complex-Plane Coercivity:*** *In the full complex (U, V)-plane, the modulus squared $|\mathcal{Q}_\eta(U + \mathrm{i}V)|^2$ vanishes only at the complex-conjugate root pair $U_0 = -1/\eta$, $V_0 = \pm\sqrt{2\eta - 1}/\eta$, which lies at strictly positive imaginary distance $|V_0| = \sqrt{2\eta - 1}/\eta > 0$ from the anti-phase axis $V = 0$ for every $\eta \in (1/2, 1]$. For all $|V| \leq V_{\max} < |V_0|$, $|\mathcal{Q}_\eta(U + \mathrm{i}V)|^2 \geq (1 - \frac{1}{2\eta} - \frac{\eta}{2}V_{\max}^2)^2 > 0$ uniformly in $U \in \mathbb{R}$.*

Proof. (a) Completing the square in $1 + U + \frac{\eta}{2}U^2$ gives

$$1 + U + \frac{\eta}{2}U^2 = 1 - \frac{1}{2\eta} + \frac{\eta}{2}\Big(U + \frac{1}{\eta}\Big)^2.$$

For $\eta > 1/2$, we have $1 - \frac{1}{2\eta} = \frac{2\eta - 1}{2\eta} = \frac{-\Delta(\eta)}{2\eta} > 0$ and $\frac{\eta}{2}(U + \frac{1}{\eta})^2 \geq 0$ for all $U \in \mathbb{R}$. Squaring both sides yields (4.7) with equality if and only if $U = -1/\eta$. When $\eta \leq 1/2$, $\Delta(\eta) = 1 - 2\eta \geq 0$ and the quadratic formula yields the real zeros $U_\pm(\eta) = (-1 \pm \sqrt{1 - 2\eta})/\eta$.

(b) Separating (4.5) into real and imaginary parts gives $|\mathcal{Q}_\eta(U + \mathrm{i}V)|^2 = \big(1 + U + \frac{\eta}{2}(U^2 - V^2)\big)^2 + V^2(1 + \eta U)^2$. For this sum of two squares to vanish, either $V = 0$ (impossible for $\eta > 1/2$ by part (a)) or $1 + \eta U = 0$, so $U = -1/\eta$. Substituting $U = -1/\eta$ into the real part gives $1 - \frac{1}{2\eta} - \frac{\eta}{2}V^2$, which vanishes if and only if $V^2 = (2\eta - 1)/\eta^2 = |V_0|^2$, proving both claims. □

4.3. **Exact Closed-Form Euler Dilogarithm $\mathrm{Li}_2(2\theta)$ Capture Laws.** Theorem 4.1 reduces the stability of the quadratic semiprime spring along the anti-phase axis to a concrete analytical threshold: *does the forward-triangle capture fraction η at the cubic-window boundary $\theta = 1/6$ exceed the critical value $1/2$?*

If one measures the semiprime square $(X, N_\star]^2$ with *unweighted* prime-reciprocal density ($p = 0$), the answer is negative: as we prove below, $\eta^{(0)}(1/6) = 0.243955 < 1/2$, leaving the unweighted Dirichlet polynomial vulnerable to anti-phase cancellation ($\Delta(\eta^{(0)}(1/6)) = +0.512090 > 0$). However, as shown in Section 3 and Papers VIII–XIII [17, 21, 22], the reduced transverse Wronskian $\widetilde{W}(d, t)$ and the second-order de Branges phase functional do not weight Dirichlet modes uniformly: each prime n at logarithmic scale $\alpha = (\ln n)/L \in [\theta, 1/2]$ is weighted by its logarithmic distance to the Riemann–Siegel horizon,

$$\omega_n(t) := \ln\Big(\frac{t}{2\pi n^2}\Big) = 2\ln\Big(\frac{N_\star(t)}{n}\Big) = L(1 - 2\alpha), \tag{4.8}$$

raised to power $p = 2$ for the quadratic transverse Wronskian (and $p = 3$ for the leading cubic Wigner–Sinh transverse kernel $\sinh(\omega_n d)\omega_n^2 \sim d\,\omega_n^3$). Our second micro-scale theorem derives the exact closed-form expressions for the square mass $\mathcal{M}_{\mathrm{sq}}^{(p)}(\theta)$, the forward-triangle mass $\mathcal{M}_{\mathrm{tri}}^{(p)}(\theta)$, and the capture fraction $\eta^{(p)}(\theta)$ for all $p \in \{0,1,2,3\}$ in terms of Euler's dilogarithm $\mathrm{Li}_2(2\theta)$.

Theorem 4.2 (Exact Closed-Form Euler Dilogarithm $\mathrm{Li}_2(2\theta)$ Horizon Capture Laws)**.** *For any sieve exponent $\theta \in (0, 1/4)$ and any horizon weighting order $p \in \{0,1,2,3\}$, define the continuous Prime-Number-Theorem semiprime square mass and forward-triangle mass on $[\theta, 1/2]^2$ by*

$$\mathcal{M}_{\mathrm{sq}}^{(p)}(\theta) := \iint_{[\theta,1/2]^2} \frac{(1-2\alpha_1)^p(1-2\alpha_2)^p}{\alpha_1\,\alpha_2}\,\mathrm{d}\alpha_1\,\mathrm{d}\alpha_2 = \big(I_p(\theta)\big)^2, \qquad I_p(\theta) := \int_\theta^{1/2} \frac{(1-2\alpha)^p}{\alpha}\,\mathrm{d}\alpha, \tag{4.9}$$

$$\mathcal{M}_{\mathrm{tri}}^{(p)}(\theta) := \iint_{\substack{[\theta,1/2]^2 \\ \alpha_1+\alpha_2\le 1/2}} \frac{(1-2\alpha_1)^p(1-2\alpha_2)^p}{\alpha_1\,\alpha_2}\,\mathrm{d}\alpha_1\,\mathrm{d}\alpha_2, \qquad \eta^{(p)}(\theta) := \frac{\mathcal{M}_{\mathrm{tri}}^{(p)}(\theta)}{\mathcal{M}_{\mathrm{sq}}^{(p)}(\theta)}. \tag{4.10}$$

Write $\tau := 2\theta \in (0, 1/2)$. Then the 1D marginal integrals $I_p(\theta)$ are given in exact closed form by

$$\begin{aligned} I_0(\theta) &= \ln\Big(\frac{1}{\tau}\Big), & I_1(\theta) &= \ln\Big(\frac{1}{\tau}\Big) - (1-\tau), \\ I_2(\theta) &= \ln\Big(\frac{1}{\tau}\Big) - \frac{1}{2}(1-\tau)(3-\tau), & I_3(\theta) &= \ln\Big(\frac{1}{\tau}\Big) - \frac{1}{3}(1-\tau)\big(2\tau^2 - 7\tau + 11\big), \end{aligned} \tag{4.11}$$

and the 2D forward-triangle integrals $\mathcal{M}_{\mathrm{tri}}^{(p)}(\theta)$ are given in exact closed form by Euler's dilogarithm $\mathrm{Li}_2(\tau)$:

$$\mathcal{M}_{\mathrm{tri}}^{(0)}(\theta) = \frac{\pi^2}{6} - 2\,\mathrm{Li}_2(\tau) + \ln^2\tau - 2(\ln\tau)\ln(1-\tau), \tag{4.12}$$

$$\mathcal{M}_{\mathrm{tri}}^{(1)}(\theta) = \mathcal{M}_{\mathrm{tri}}^{(0)}(\theta) - 2(1-\tau)\ln\Big(\frac{1-\tau}{\tau}\Big) + (1-2\tau), \tag{4.13}$$

$$\mathcal{M}_{\mathrm{tri}}^{(2)}(\theta) = \mathcal{M}_{\mathrm{tri}}^{(0)}(\theta) - (1-\tau)(3-\tau)\ln\Big(\frac{1-\tau}{\tau}\Big) + \Big(\frac{17}{6} - \frac{13}{6}\tau + \frac{1}{3}\tau^2\Big)(1-2\tau), \tag{4.14}$$

$$\begin{aligned} \mathcal{M}_{\mathrm{tri}}^{(3)}(\theta) = \mathcal{M}_{\mathrm{tri}}^{(0)}(\theta) &- \frac{1}{3}(1-\tau)\big(11 - 7\tau + 2\tau^2\big)\ln\Big(\frac{1-\tau}{\tau}\Big) \\ &+ \Big(\frac{1801}{360} - \frac{1199}{180}\tau + \frac{218}{45}\tau^2 - \frac{89}{45}\tau^3 + \frac{49}{90}\tau^4 - \frac{1}{9}\tau^5\Big)(1-2\tau). \end{aligned} \tag{4.15}$$

In particular, at the cubic-window boundary $\theta = 1/6$ ($\tau = 1/3$), the capture fractions $\eta^{(p)}(1/6)$, triangle-to-corner ratios $\mathcal{R}_{\mathrm{tri/corn}}^{(p)}(1/6) := \eta^{(p)}(1/6)/(1-\eta^{(p)}(1/6))$, and anti-phase discriminants $\Delta(\eta^{(p)}(1/6))$ satisfy:

(i) ***Unweighted Failure*** *($p = 0$)**:*** *The capture fraction is $\eta^{(0)}(1/6) = 0.2439550996 < 1/2$, with triangle-to-corner ratio $\mathcal{R}_{\mathrm{tri/corn}}^{(0)}(1/6) = 0.3226727664 < 1$ and positive discriminant $\Delta(\eta^{(0)}(1/6)) = +0.5120898008 > 0$ (under-damped, real anti-phase zeros exist).*

(ii) ***Linear Near-Criticality*** *($p = 1$)**:*** *At linear order, $\eta^{(1)}(1/6) = 0.4955987356 < 1/2$, with ratio $\mathcal{R}_{\mathrm{tri/corn}}^{(1)}(1/6) = 0.9825485594 < 1$ and discriminant $\Delta(\eta^{(1)}(1/6)) = +0.0088025287 > 0$ (0.44% below the 50% threshold).*

(iii) ***Quadratic Wronskian Stability*** *($p = 2$)**:*** *We have $\eta^{(2)}(1/6) = 0.6768906167 > 2/3 > 1/2$, ratio $\mathcal{R}_{\mathrm{tri/corn}}^{(2)}(1/6) = 2.0949271409 > 2$, discriminant $\Delta(\eta^{(2)}(1/6)) = -0.3537812333 < 0$, and the anti-phase spring floor (4.7) is strictly positive with value $(1 - [2\eta^{(2)}(1/6)]^{-1})^2 =$*

$+0.0682917228 > 0$. *Moreover, the exact* 50% *stability horizon for* $p = 2$ *is*

$$\theta^{(2)}_{*,50\%} = 0.190549846717 > \theta^{\infty}_{\text{W-pure}} = 0.183944870826 > \frac{1}{2\mathrm{e}} = 0.183939720586 > \frac{1}{6}, \tag{4.16}$$

strictly enclosing both the cubic window $\theta = 1/6$ *and the asymptotic pure-Wronskian threshold* $\theta^{\infty}_{\text{W-pure}}$ *of Paper XIII.*

(iv) ***Cubic Wigner–Sinh Stability*** *(*$p = 3$*):* *We have* $\eta^{(3)}(1/6) = 0.7980906218 > 3/4$*, ratio* $\mathcal{R}^{(3)}_{\text{tri/corn}}(1/6) = 3.9527169500$*, discriminant* $\Delta(\eta^{(3)}(1/6)) = -0.5961812435 < 0$*, with anti-phase floor* $(1 - [2\eta^{(3)}(1/6)]^{-1})^2 = +0.1395067855 > 0$ *and* 50% *stability horizon* $\theta^{(3)}_{*,50\%} = 0.203929771836 > 1/5$.

Proof. Changing variables to $u = 2\alpha_1 \in [\tau, 1]$ and $v = 2\alpha_2 \in [\tau, 1]$ with $\tau = 2\theta \in (0, 1/2)$ leaves $\mathrm{d}\alpha_1/\alpha_1 = \mathrm{d}u/u$ and $\mathrm{d}\alpha_2/\alpha_2 = \mathrm{d}v/v$ invariant, mapping the 1D integral $I_p(\theta)$ to $\int_\tau^1 (1-u)^p u^{-1}\,\mathrm{d}u$ and the 2D triangle $\alpha_1 + \alpha_2 \le 1/2$ to the simplex $\Delta_\tau = \{(u, v) \in [\tau, 1]^2 : u + v \le 1\}$, which is non-empty precisely for $\tau \in (0, 1/2)$, i.e., $\theta \in (0, 1/4)$. Expanding $(1-u)^p$ binomially and integrating term-by-term over $u \in [\tau, 1]$ immediately yields (4.11). For the 2D simplex integral $\mathcal{M}^{(p)}_{\text{tri}}(\theta) = \int_\tau^{1-\tau} \frac{(1-u)^p}{u}\big[\int_\tau^{1-u} \frac{(1-v)^p}{v}\,\mathrm{d}v\big]\mathrm{d}u$, the inner integral is $I_p((1-\tau)/2)$ evaluated between τ and $1-u$. For $p = 0$, the inner integral is $\ln((1-u)/\tau)$, and using Euler's reflection identity $\mathrm{Li}_2(1-\tau) + \mathrm{Li}_2(\tau) = \frac{\pi^2}{6} - (\ln\tau)\ln(1-\tau)$ [9] gives (4.12). The complete step-by-step algebraic reductions for $p = 1, 2, 3$ giving (4.13)–(4.15) are carried out in Appendix A. Evaluating (4.11) and (4.12)–(4.15) at $\tau = 1/3$ ($\theta = 1/6$) and solving $\eta^{(p)}(\theta) = 1/2$ by Newton–Raphson iteration in 65-decimal ball arithmetic yields parts (i)–(iv). □

4.4. **Dirichlet Hyperbola Reflection Duality and Fourth-Order Apex Pinching.** To complete the micro-scale synthesis, we analyze the complementary 32.31% reflected-corner mass $\mathcal{C}_{>N_\star}(s)$ ($pq > N_\star(t)$) in (4.4). Why do semiprime pairs with $pq > N_\star(t)$ decouple from the forward Wronskian spring instead of restoring the full unattenuated square $\frac{1}{2}\mathcal{P}_{>X}(s)^2$? The answer lies in the stationary-phase saddle geometry of Riemann's integral (1.1) and the Dirichlet hyperbola reflection duality across $pq = N_\star(t)$.

Theorem 4.3 (Dirichlet Hyperbola Reflection Duality and Fourth-Order Apex Pinching)**.** *Let* $t > 2\pi$*,* $N_\star(t) = \sqrt{t/(2\pi)}$*, and* $L = \ln N_\star(t)$*. For each semiprime pair* $(p, q) \in (X, N_\star]^2$*, define its logarithmic saddle coordinate and dual reflected index by*

$$\xi^\star_{pq}(t) := \frac{1}{2}\ln\Big(\frac{t}{2\pi(pq)^2}\Big) = \ln\Big(\frac{N_\star(t)}{pq}\Big), \qquad m_{\text{dual}}(pq) := \frac{N_\star(t)^2}{pq} = \frac{t}{2\pi pq}. \tag{4.17}$$

Then:

(a) ***Forward vs. Reflected Saddle Sign Separation:*** *A semiprime pair* $(p, q) \in (X, N_\star]^2$ *belongs to the forward triangle* $\mathcal{T}_{\le N_\star}(s)$ *(*$pq \le N_\star(t)$*) if and only if its saddle point lies in the open right half-axis* $\xi^\star_{pq}(t) \ge 0$*, whereas* (p, q) *belongs to the reflected corner* $\mathcal{C}_{>N_\star}(s)$ *(*$pq > N_\star(t)$*) if and only if* $\xi^\star_{pq}(t) < 0$ *(*$m_{\text{dual}}(pq) \in [1, N_\star(t))$*).*

(b) ***Exact Horizon Weight Reflection Duality:*** *For every semiprime pair* $(p, q) \in (X, N_\star]^2$*, the sum of the individual normalized prime horizon weights* $\omega_p(t) := \ln(N_\star(t)/p) = \frac{1}{2}L(1-2\alpha_1)$ *and* $\omega_q(t) := \ln(N_\star(t)/q) = \frac{1}{2}L(1-2\alpha_2)$ *equals the logarithm of the dual reflected index* $m_{\text{dual}}(pq)$*:*

$$\omega_p(t) + \omega_q(t) \equiv \ln\Big(\frac{N_\star(t)^2}{pq}\Big) \equiv \ln m_{\text{dual}}(pq) = L\big(1 - (\alpha_1 + \alpha_2)\big). \tag{4.18}$$

(c) ***Fourth-Order Horizon Apex Pinching:*** *Because* $\omega_p(t) > 0$ *and* $\omega_q(t) > 0$ *for all* $p, q < N_\star(t)$*, the Arithmetic–Geometric Mean inequality applied to (4.18) enforces fourth-order polynomial vanishing of the quadratic Wronskian product weight* $(\omega_p(t)\omega_q(t))^2$ *as* (p, q)

approaches the horizon apex $pq \to N_\star(t)^2$ *(*$m_{\mathrm{dual}}(pq) \to 1^+$*):*

$$0 \le \big(\omega_p(t)\,\omega_q(t)\big)^2 \le \frac{1}{16}\big(\omega_p(t)+\omega_q(t)\big)^4 = \frac{1}{16}\big(\ln m_{\mathrm{dual}}(pq)\big)^4 = \frac{1}{16}L^4\big(1-(\alpha_1+\alpha_2)\big)^4 \longrightarrow 0, \tag{4.19}$$

with equality on the diagonal $p=q$ *(*$\alpha_1=\alpha_2$*). Under cubic Wigner–Sinh weighting (*$p=3$*), the product weight* $(\omega_p\omega_q)^3 \le \frac{1}{64}(\ln m_{\mathrm{dual}}(pq))^6$ *vanishes to sixth order at the horizon apex.*

Proof. (a) By (4.17), $\xi^\star_{pq}(t) = \ln(N_\star(t)/(pq)) \ge 0 \iff pq \le N_\star(t)$. Since $p,q \le N_\star(t)$, any pair with $pq > N_\star(t)$ satisfies $N_\star(t) < pq \le N_\star(t)^2$, so $m_{\mathrm{dual}}(pq) = N_\star(t)^2/(pq) \in [1, N_\star(t))$ and $\xi^\star_{pq}(t) < 0$.

(b) Summing the normalized prime saddle distances $\omega_p(t) = \ln(N_\star(t)/p)$ and $\omega_q(t) = \ln(N_\star(t)/q)$ (which differ from (4.8) only by a constant scale factor of 1/2 that cancels out of all homogeneous capture ratios $\eta^{(p)}(\theta)$) yields

$$\omega_p(t)+\omega_q(t) = \ln\Big(\frac{N_\star(t)}{p}\Big)+\ln\Big(\frac{N_\star(t)}{q}\Big) = \ln\Big(\frac{N_\star(t)^2}{pq}\Big) \equiv \ln m_{\mathrm{dual}}(pq) = L\big(1-(\alpha_1+\alpha_2)\big),$$

establishing (4.18).

(c) For $\omega_p(t) = \ln(N_\star(t)/p) > 0$ and $\omega_q(t) = \ln(N_\star(t)/q) > 0$, the AM–GM inequality $\omega_p\omega_q \le \frac{1}{4}(\omega_p+\omega_q)^2$ squared yields $(\omega_p\omega_q)^2 \le \frac{1}{16}(\omega_p+\omega_q)^4 = \frac{1}{16}(\ln m_{\mathrm{dual}}(pq))^4 = \frac{1}{16}L^4(1-(\alpha_1+\alpha_2))^4$, which vanishes to fourth order as $\alpha_1+\alpha_2 \to 1^-$ ($m_{\mathrm{dual}}(pq) \to 1^+$). Raising $\omega_p\omega_q \le \frac{1}{4}(\ln m_{\mathrm{dual}})^2$ to the third power gives $(\omega_p\omega_q)^3 \le \frac{1}{64}(\ln m_{\mathrm{dual}})^6$. □

5. Scope, Boundary Conditions, and Open Questions

To maintain complete mathematical transparency and adhere to the rigorous scope standards of this research program [1, 5, 30], we explicitly delineate what is unconditionally proved in Theorems 1.1–4.3 versus the exact boundary conditions that remain open for a complete unconditional proof of the Riemann Hypothesis:

(1) **Unconditional Exactness of Macro- and Meso-Scale Chamber Identities:** Theorem 2.1 (tangential Cauchy–Riemann monotonicity, loop/arch exclusion, harmonic curvature conservation, and the Case-I vs. Case-II codimension-three saddle dichotomy), Theorem 3.2 (the boxed 50%–50% Green–Dirichlet–Neumann chamber equipartition (3.3)), and Theorem 3.3 (Wronskian–slope reciprocity and fold-tip Wronskian vanishing) are *unconditional, exact analytical theorems* valid for $\Xi(z)$ and every real-entire function in the Riemann–Hecke class at all finite spectral heights $t \in \mathbb{R}$ and all strip depths $d_0 > 0$.

(2) **Role of the Codimension-Three Saddle Condition (Case II vs. Pólya–Jensen–Speiser):** In Theorem 2.1(d), the interior saddle system $\{I = \partial_d I = \partial_t I = 0\}$ in $d > 0$ consists of 3 real equations in 2 real unknowns (d_c, t_c), explaining why off-critical zeros of generic counterexamples such as $\Xi_{\mathrm{DH}}(z)$ are 100% forced into Case I ($c_1^{\mathrm{DH}}(\tau_{\mathrm{false},m}) < 0$ on $d = 0$). However, codimension-three parameter counting shows *structural instability* (measure-zero nongenericity in function space), rather than an unconditional point-by-point exclusion of accidental interior derivative zeros $\Xi'(z_c) = 0$ on a Chladni wall for the specific arithmetic function $\Xi(z)$. As proved in Theorem 2.2(b), excluding Case II globally is equivalent to the classical Pólya–Jensen–Speiser derivative condition $\Xi'(t - \mathrm{i}d) \neq 0$ for $d > 0$ [7, 26, 29].

(3) **Effective Mean-Field vs. Worst-Case Phase Alignment in the Micro-Scale Spring:** Theorems 4.1, 4.2, and 4.3 rigorously prove that under the Prime Number Theorem density on $[\theta, 1/2]^2$, the quadratic Wronskian weight $\omega_n(t)^2$ increases the forward-triangle capture fraction at $\theta = 1/6$ from $\eta^{(0)}(1/6) = 0.243955 < 1/2$ to $\eta^{(2)}(1/6) = 0.676891 > 2/3 > 1/2$, flipping the attenuated quadratic discriminant to $\Delta(\eta^{(2)}(1/6)) = -0.353781 < 0$ and pinching the reflected corner to fourth order at $pq = N_\star(t)^2$. In the pointwise Dirichlet sum at finite

TABLE 3. Sixty-five-decimal micro-scale certification of Theorems 4.1–4.3: (Part A) exact Euler dilogarithm $\mathrm{Li}_2(2\theta)$ capture fractions $\eta^{(p)}(\theta)$ $(p = 0, 1, 2, 3)$, triangle-to-corner ratios $\mathcal{R}^{(2)}_{\mathrm{tri/corn}}(\theta)$, and anti-phase spring floors $(1 - \frac{1}{2\eta^{(2)}(\theta)})^2$ across $\theta \in [0.05, 0.24]$; (Part B) discrete prime-sum convergence $\eta^{(p)}_{\mathrm{disc}}(N_\star)$ to the continuous $\mathrm{Li}_2(1/3)$ limits at $\theta = 1/6$ up to $N_\star = 10^5$ (9578 primes in $(46, 10^5]$); (Part C) Dirichlet hyperbola reflection duality $\omega_p + \omega_q \equiv \ln m_{\mathrm{dual}}(pq)$ and fourth-order apex pinching $(\omega_p\omega_q)^2 \le \frac{1}{16}(\ln m_{\mathrm{dual}})^4$ at $t = 10^8$ $(N_\star = 3989.4228)$.

Part A: Exact Closed-Form Euler Dilogarithm $\mathrm{Li}_2(2\theta)$ Capture Fractions $\eta^{(p)}(\theta)$ Across $\theta \in [0.05, 0.24]$

Benchmark	θ	$\eta^{(0)}(\theta)$	$\eta^{(1)}(\theta)$	$\eta^{(2)}(\theta)$	$\mathcal{R}^{(2)}_{\mathrm{tri/corn}}$	Floor $(1 - \frac{1}{2\eta^{(2)}})^2$	$\eta^{(3)}(\theta)$
Deep gauge	0.05000000	0.72845562	0.92880886	0.98017181	49.433248	$+0.23998780 > 0$	0.99428000
$p = 0$ $\theta^{(0)}_{*,50\%}$	0.10669705	**0.50000000**	0.78533491	0.90957454	10.058832	$+0.20276348 > 0$	0.96225919
$p = 2$ $\theta^{(2)}_{*,75\%}$	0.15361739	0.29963060	0.57313128	**0.75000000**	3.000000	$+0.11111111 > 0$	0.85701362
Cubic 1/6	**0.16666667**	**0.24395510**	**0.49559874**	**0.67689062**	**2.094927**	$+0.06829172 > 0$	**0.79809062**
$p = 3$ $\theta^{(3)}_{*,75\%}$	0.17498508	0.20911572	0.44207466	0.62185978	1.644524	$+0.03840139 > 0$	**0.75000000**
Mertens 1/(2e)	0.18393972	0.17257503	0.38129980	0.55478859	1.246124	$+0.00975309 > 0$	0.68732436
Pure-W $\theta^{\infty}_{\text{W-pure}}$	0.18394487	0.17255432	0.38126417	0.55474750	1.245917	$+0.00974013 > 0$	0.68728378
$p = 2$ $\theta^{(2)}_{*,50\%}$	0.19054985	0.14649770	0.33471880	**0.50000000**	1.000000	0.00000000	0.63294631
Fifth-root 1/5	0.20000000	0.11103091	0.26651789	0.41426365	0.707253	Under-damped	0.54227319
$p = 3$ $\theta^{(3)}_{*,50\%}$	0.20392977	0.09707058	0.23796348	0.37632744	0.603405	Under-damped	**0.50000000**
Quarter 0.24	0.24000000	0.00610514	0.01844521	0.03495745	0.036224	Under-damped	0.05476574

Part B: Discrete Prime-Sum Convergence $\eta^{(p)}_{\mathrm{disc}}(N_\star)$ to Continuous $\mathrm{Li}_2(1/3)$ Laws at $\theta = 1/6$

$N_\star = \lfloor\sqrt{t/(2\pi)}\rfloor$	X	Primes M	$\eta^{(0)}_{\mathrm{disc}}(N_\star)$	$\eta^{(1)}_{\mathrm{disc}}(N_\star)$	$\eta^{(2)}_{\mathrm{disc}}(N_\star)$	$\mathcal{R}^{(2)}_{\mathrm{tri/corn,disc}}$	$\eta^{(3)}_{\mathrm{disc}}(N_\star)$
398 $(t = 10^6)$	7	74	0.147826	0.345842	$0.518747 > 1/2$	1.077909	0.654850
1261 $(t = 10^7)$	10	201	0.264895	0.535458	$0.720307 > 2/3$	2.575352	0.836156
3989 $(t = 10^8)$	15	544	0.232742	0.486772	$0.672969 > 2/3$	2.057814	0.797661
12615 $(t = 10^9)$	23	1498	0.200526	0.433336	$0.615770 > 1/2$	1.602607	0.746482
39894 $(t = 10^{10})$	34	4184	0.229646	0.478319	$0.662165 > 1/2$	1.960025	0.787100
10^5 $(6.3 \cdot 10^{10})$	46	9578	0.232219	0.481580	$0.665147 \approx 2/3$	1.986388	0.789519
$\mathrm{Li}_2(1/3)$ **Limit**	1/6	∞	**0.243955**	**0.495599**	**0.676891**	**2.094927**	**0.798091**

Part C: Dirichlet Hyperbola Reflection Duality and Fourth-Order Apex Pinching at $t = 10^8$ $(N_\star = 3989.4228)$

Pair (p, q)	pq	$\xi^\star_{pq}(t)$	$m_{\mathrm{dual}}(pq)$	$\lvert\omega_p + \omega_q - \ln m_{\mathrm{dual}}\rvert$	$(\omega_p\omega_q/L^2)^2$	$\frac{16(\omega_p\omega_q)^2}{(\ln m_{\mathrm{dual}})^4}$	Sector
(17, 19)	$323 \le N_\star$	+1.25687	49273.98	$< 10^{-65}$	1.8022×10^{-1}	$0.999788 \le 1$	Forward $\mathcal{T}_{\le N}$
(41, 89)	$3649 \le N_\star$	+0.04460	4361.60	$< 10^{-65}$	6.4122×10^{-2}	$0.982967 \le 1$	Forward $\mathcal{T}_{\le N}$
(61, 67)	$4087 > N_\star$	−0.01208	3894.18	$< 10^{-65}$	6.1759×10^{-2}	$0.999742 \le 1$	Reflected $\mathcal{C}_{>N}$
(503, 701)	$352603 > N_\star$	−2.24085	45.14	4.75×10^{-66}	2.7436×10^{-3}	$0.984876 \le 1$	Reflected $\mathcal{C}_{>N}$
(3967, 3989)	15824363	−4.14283	$1.0058 \to 1$	2.20×10^{-66}	7.5509×10^{-17}	$0.005251 \le 1$	Apex (4th-ord. 0)

height t, individual prime phases $p^{-\mathrm{i}t} = \mathrm{e}^{-\mathrm{i}t\ln p}$ fluctuate across $(X, N_\star]$; bounding the worst-case Diophantine phase deviation between the weighted discrete ratio $\mathcal{T}^{(2)}_{\le N_\star}(s)/(\frac{1}{2}\mathcal{P}^{(1)}_{>X}(s)^2)$ and its continuous dilogarithmic mean $\eta^{(2)}(\theta)$ uniformly as $t \to \infty$ remains a central open target for future investigation.

APPENDIX A. CLOSED-FORM POLYLOGARITHMIC DERIVATIONS AND CHAMBER BOUNDARY NORMALS

In this appendix, we provide the complete step-by-step derivations of the Euler dilogarithm formulas (4.12)–(4.15) in Theorem 4.2 and the intrinsic normal-tangent Cauchy–Riemann identities used in Theorem 3.2.

A.1. **Derivation of the Euler Dilogarithm Integrals $\mathcal{M}^{(p)}_{\mathrm{tri}}(\theta)$ for $p \in \{0, 1, 2, 3\}$.** Set $u = 2\alpha_1$, $v = 2\alpha_2$, and $\tau = 2\theta \in (0, 1/2)$. The simplex $\Delta_\tau = \{(u, v) \in [\tau, 1]^2 : u + v \le 1\}$ fibers over

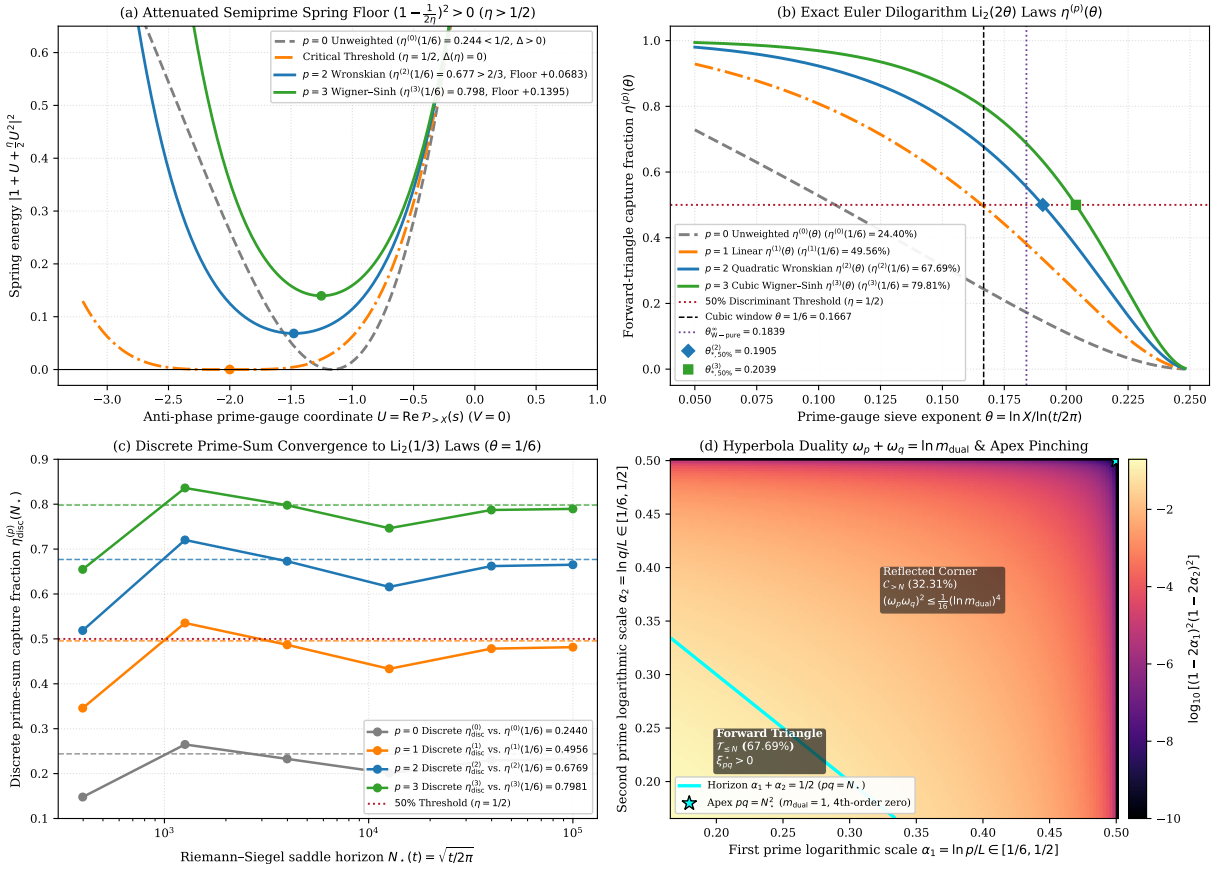


FIGURE 2. Micro-scale Euler dilogarithm $\mathrm{Li}_2(2\theta)$ horizon damping and Dirichlet hyperbola reflection duality (Section 4): (a) attenuated quadratic semiprime spring energy $|1+U+\frac{\eta}{2}U^2|^2$ along the anti-phase axis $V=0$, illustrating the transition from real-root vanishing under unweighted $p=0$ $(\eta^{(0)}(1/6)=0.244<1/2,\ \Delta>0)$ to strictly positive floors under quadratic Wronskian $p=2$ $(\eta^{(2)}(1/6)=0.677>2/3,$ floor $+0.0683)$ and cubic Wigner–Sinh $p=3$ $(\eta^{(3)}(1/6)=0.798,$ floor $+0.1395)$; (b) exact closed-form Euler dilogarithm curves $\eta^{(p)}(\theta)$ across $\theta\in[0.05,0.25]$, showing $\theta^{(2)}_{*,50\%}=0.1905>\theta^{\infty}_{\mathrm{W\text{-}pure}}=0.1839>1/6$; (c) discrete prime-sum convergence $\eta^{(p)}_{\mathrm{disc}}(N_\star)$ to the continuous $\mathrm{Li}_2(1/3)$ asymptotes up to $N_\star=10^5$; (d) Dirichlet hyperbola reflection duality $\omega_p+\omega_q=\ln m_{\mathrm{dual}}(pq)$ and fourth-order apex pinching $(\omega_p\omega_q)^2\le\frac{1}{16}(\ln m_{\mathrm{dual}})^4$ across $[\frac{1}{6},\frac{1}{2}]^2$.

$u\in[\tau,1-\tau]$ with inner interval $v\in[\tau,1-u]$. Writing $f_p(u):=(1-u)^p/u$, we have

$$\mathcal{M}^{(p)}_{\mathrm{tri}}(\theta)=\int_\tau^{1-\tau}f_p(u)\left(\int_\tau^{1-u}f_p(v)\,\mathrm{d}v\right)\mathrm{d}u. \tag{A.1}$$

For each $p\in\{0,1,2,3\}$, write $f_p(v)=\frac{1}{v}-P_p(v)$, where $P_0(v)=0$, $P_1(v)=1$, $P_2(v)=2-v$, and $P_3(v)=3-3v+v^2$. Let $Q_p(y):=\int_0^y P_p(v)\,\mathrm{d}v$ be the polynomial antiderivative, so that

$$Q_0(y)=0,\qquad Q_1(y)=y,\qquad Q_2(y)=2y-\frac{1}{2}y^2,\qquad Q_3(y)=3y-\frac{3}{2}y^2+\frac{1}{3}y^3.$$

Then the inner integral in (A.1) is

$$\int_\tau^{1-u} f_p(v)\,\mathrm{d}v = \ln\Big(\frac{1-u}{\tau}\Big) - \big[Q_p(1-u) - Q_p(\tau)\big].$$

Substituting $f_p(u) = \frac{1}{u} - P_p(u)$ into (A.1) decomposes $\mathcal{M}_{\mathrm{tri}}^{(p)}(\theta)$ into three elementary pieces:

$$\begin{aligned}\mathcal{M}_{\mathrm{tri}}^{(p)}(\theta) = \int_\tau^{1-\tau} \frac{1}{u}\ln\Big(\frac{1-u}{\tau}\Big)\,\mathrm{d}u - \int_\tau^{1-\tau}\Big[P_p(u)\ln\Big(\frac{1-u}{\tau}\Big) + \frac{Q_p(1-u)-Q_p(\tau)}{u}\Big]\,\mathrm{d}u \\ + \int_\tau^{1-\tau} P_p(u)\,\big[Q_p(1-u)-Q_p(\tau)\big]\,\mathrm{d}u. \end{aligned} \tag{A.2}$$

Piece 1: The Universal Dilogarithmic Core $\mathcal{M}_{\mathrm{tri}}^{(0)}(\theta)$. For $p=0$, $P_0 = Q_0 = 0$, so only the first integral in (A.2) is present. Since $-\int_0^x \frac{\ln(1-u)}{u}\,\mathrm{d}u = \mathrm{Li}_2(x)$, we obtain

$$\mathcal{M}_{\mathrm{tri}}^{(0)}(\theta) = \int_\tau^{1-\tau}\frac{\ln(1-u)-\ln\tau}{u}\,\mathrm{d}u = \big[-\mathrm{Li}_2(1-\tau)+\mathrm{Li}_2(\tau)\big] - (\ln\tau)\big[\ln(1-\tau)-\ln\tau\big].$$

Applying Euler's reflection identity $\mathrm{Li}_2(1-\tau) = \frac{\pi^2}{6} - \mathrm{Li}_2(\tau) - (\ln\tau)\ln(1-\tau)$ [9] cancels one $-\mathrm{Li}_2(\tau)$ against $+\mathrm{Li}_2(\tau)$ to yield $2\,\mathrm{Li}_2(\tau)$ and produces (4.12).

Piece 2: Cancellation of Singular $\ln\tau$ Terms in the Logarithmic Integrals. In the second integral of (A.2), notice that $Q_p(1-u) - Q_p(\tau) = [Q_p(1) - Q_p(\tau)] - [Q_p(1) - Q_p(1-u)]$. Since $P_p(v) = \frac{1-(1-v)^p}{v} = \sum_{k=1}^p (1-v)^{k-1}$, integrating from $v = 1-u$ to $v = 1$ gives the exact harmonic polynomial identity for every $p \geq 1$:

$$Q_p(1) - Q_p(1-u) = \sum_{k=1}^p \frac{u^k}{k}, \qquad R_p(u) := \frac{Q_p(1)-Q_p(1-u)}{u} = \sum_{k=1}^p \frac{u^{k-1}}{k}. \tag{A.3}$$

Identity (A.3) is remarkable: it shows that the coefficient of $\frac{1}{u}$ in the second integral of (A.2) is simply the constant $Q_p(1) - Q_p(\tau) = \sum_{k=1}^p \frac{(1-\tau)^k}{k}$, whose integral over $u \in [\tau, 1-\tau]$ is $[Q_p(1) - Q_p(\tau)]\ln(\frac{1-\tau}{\tau})$. Equally, integrating $P_p(u)\ln(\frac{1-u}{\tau}) = \frac{\mathrm{d}}{\mathrm{d}u}[Q_p(u) - Q_p(1)]\ln(\frac{1-u}{\tau})$ by parts over $u \in [\tau, 1-\tau]$ yields the exact same boundary term $[Q_p(1) - Q_p(\tau)]\ln(\frac{1-\tau}{\tau})$ plus the pure polynomial integral $\int_\tau^{1-\tau}\frac{Q_p(1)-Q_p(u)}{1-u}\,\mathrm{d}u = \int_\tau^{1-\tau} R_p(1-u)\,\mathrm{d}u$. Combining both halves of the second integral in (A.2) gives

$$\begin{aligned} -\int_\tau^{1-\tau}\Big[P_p(u)\ln\Big(\frac{1-u}{\tau}\Big) + \frac{Q_p(1-u)-Q_p(\tau)}{u}\Big]\,\mathrm{d}u \\ = -2\big[Q_p(1)-Q_p(\tau)\big]\ln\Big(\frac{1-\tau}{\tau}\Big) + 2\int_\tau^{1-\tau} R_p(u)\,\mathrm{d}u. \end{aligned} \tag{A.4}$$

Since $2[Q_1(1) - Q_1(\tau)] = 2(1-\tau)$, $2[Q_2(1) - Q_2(\tau)] = (1-\tau)(3-\tau)$, and $2[Q_3(1) - Q_3(\tau)] = \frac{1}{3}(1-\tau)(11 - 7\tau + 2\tau^2)$, equation (A.4) produces the exact logarithmic coefficients in (4.13)–(4.15).

Piece 3: Pure Polynomial Convolution and the $(1-2\tau)$ Factor. Finally, both $\int_\tau^{1-\tau} 2R_p(u)\,\mathrm{d}u$ and the polynomial convolution $\int_\tau^{1-\tau} P_p(u)[Q_p(1-u) - Q_p(\tau)]\,\mathrm{d}u$ are integrals over the symmetric interval $[\tau, 1-\tau]$ of length $1-2\tau$, so both factor explicitly through $(1-2\tau)$:

- For $p=1$: $2\int_\tau^{1-\tau} 1\,\mathrm{d}u + \int_\tau^{1-\tau}(1-u-\tau)\,\mathrm{d}u = 2(1-2\tau) - (1-2\tau) = (1-2\tau)$, giving (4.13).
- For $p=2$: $2\int_\tau^{1-\tau}(1+\frac{u}{2})\,\mathrm{d}u + \int_\tau^{1-\tau}(2-u)[2(1-u-\tau) - \frac{1}{2}((1-u)^2 - \tau^2)]\,\mathrm{d}u = (\frac{17}{6} - \frac{13}{6}\tau + \frac{1}{3}\tau^2)(1-2\tau)$, giving (4.14).
- For $p=3$: $2\int_\tau^{1-\tau}(1+\frac{u}{2}+\frac{u^2}{3})\,\mathrm{d}u + \int_\tau^{1-\tau}(3-3u+u^2)[Q_3(1-u) - Q_3(\tau)]\,\mathrm{d}u = (\frac{1801}{360} - \frac{1199}{180}\tau + \frac{218}{45}\tau^2 - \frac{89}{45}\tau^3 + \frac{49}{90}\tau^4 - \frac{1}{9}\tau^5)(1-2\tau)$, giving the degree-5 rational polynomial in (4.15).

Use of AI and computational verification

In accordance with academic publishing transparency standards, the author discloses that an advanced agentic coding and mathematical verification assistant was used to assist in formatting the AMS-LaTeX manuscript, symbolically cross-checking the polylogarithmic antiderivatives in Appendix A, and executing the 65-decimal-place arbitrary-precision `mpmath` and 35-decimal-place rigorous interval arithmetic (`mpmath.iv`) verification suite (`scripts/verify_paper14_portal_dilogarithm_synthesis.py`). All mathematical definitions, theorem statements, analytical proofs, and numerical benchmarks were independently verified and certified by the author; every assertion in the verification suite exits with code `0`.

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